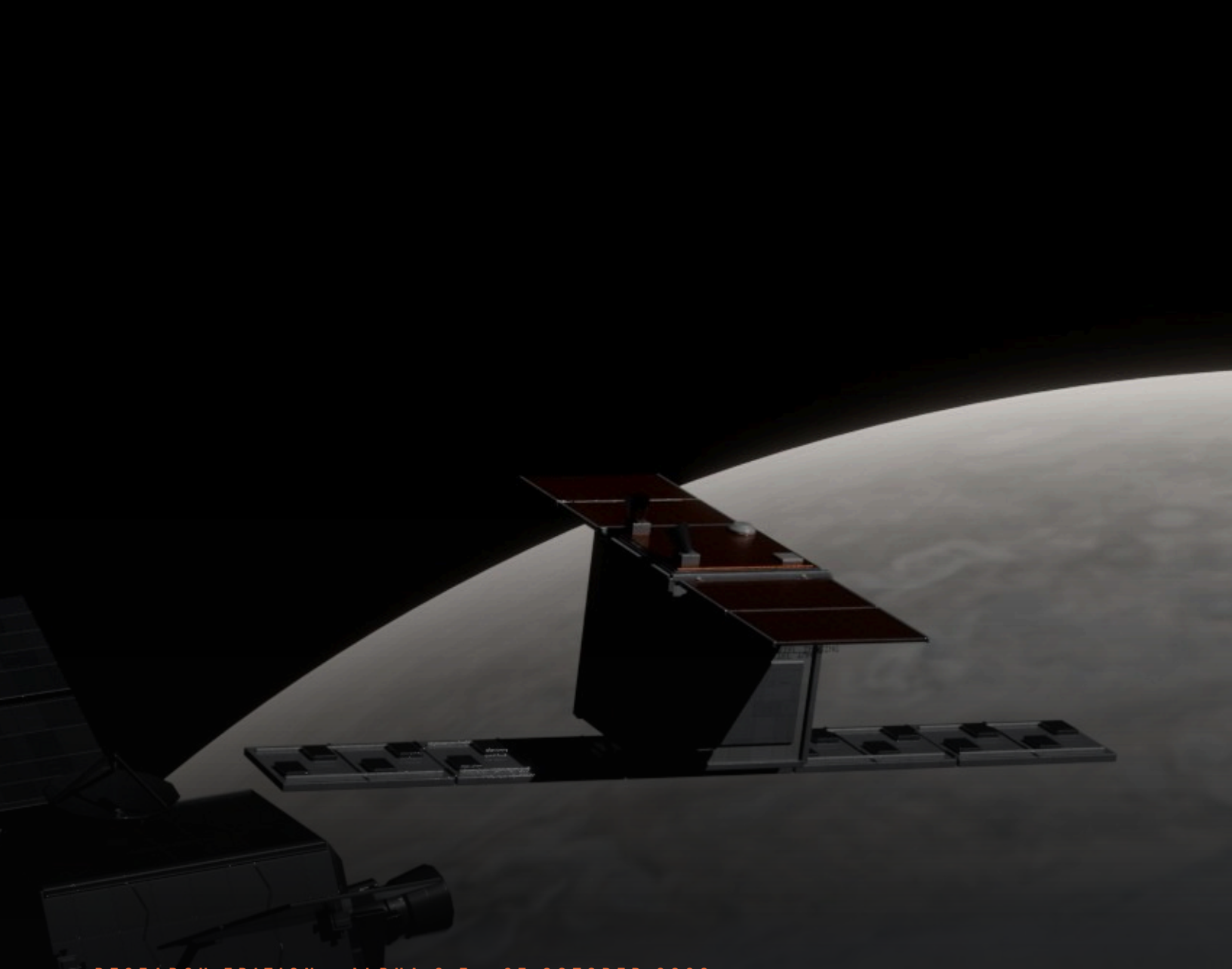


RESEARCH EDITION · ALPHA 0.5 · 25 OCTOBER 2026

xGEO
CORE

The simulation engine under xGEO: its physical models, the sources they rest on, how each is checked, and how to use it as an open-source tool

K. Williamson · xGEO project

BUILD	TESTS	COMPARISONS	ENGINE
Alpha 0.5	63	762 / 762 pass	Godot 4.7

762

of 762 comparisons within tolerance

63

automated tests against independent references

14

physical domains, from orbits to infrared

4

known simplifications, reported, not hidden

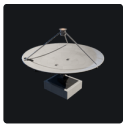
ABSTRACT

xGEO simulates a gray-zone confrontation in low Jupiter orbit in 2041: an uninvited inspector shadows an allied radar satellite, and the player has to make it leave without debris or open war. This edition sets out every physical model the game uses, with its equation, its source, how sure we can be of it, and which numbers are *design values* chosen for play. The game's own code is checked against JPL Horizons, hapsira, scipy, astropy, filterpy, ITU-R recommendations and published worked examples: 762 comparisons across 63 tests, all within tolerance, and 4 further rows that report known simplifications. Section 11 lists what could not be verified.

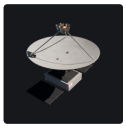
FIG. 1 WHAT THE GAME MODELS, AND WHERE TO READ ABOUT IT



ORBITS & EPHEMERIDES
§2–3 · Horizons, J2, Lambert



RADIO & DETECTION
§4 · ITU patterns, Friis, DF



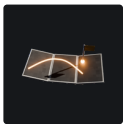
ELECTROMAGNETIC ATTACK
§5 · J/S, spot and barrage



HEAT & SIGNATURES
§6 · energy balance, IR, shadow



TRACKING
§7 · unscented Kalman filter



DOCTRINE & SCENARIO
§8–9 · JP 3-60, JP 3-85

Every model is tested against an independent reference (§10). Click a box to jump to its section.

1 METHOD AND LABELS

The rule is that every number is either sourced or labelled. Each source claim carries one of four labels:

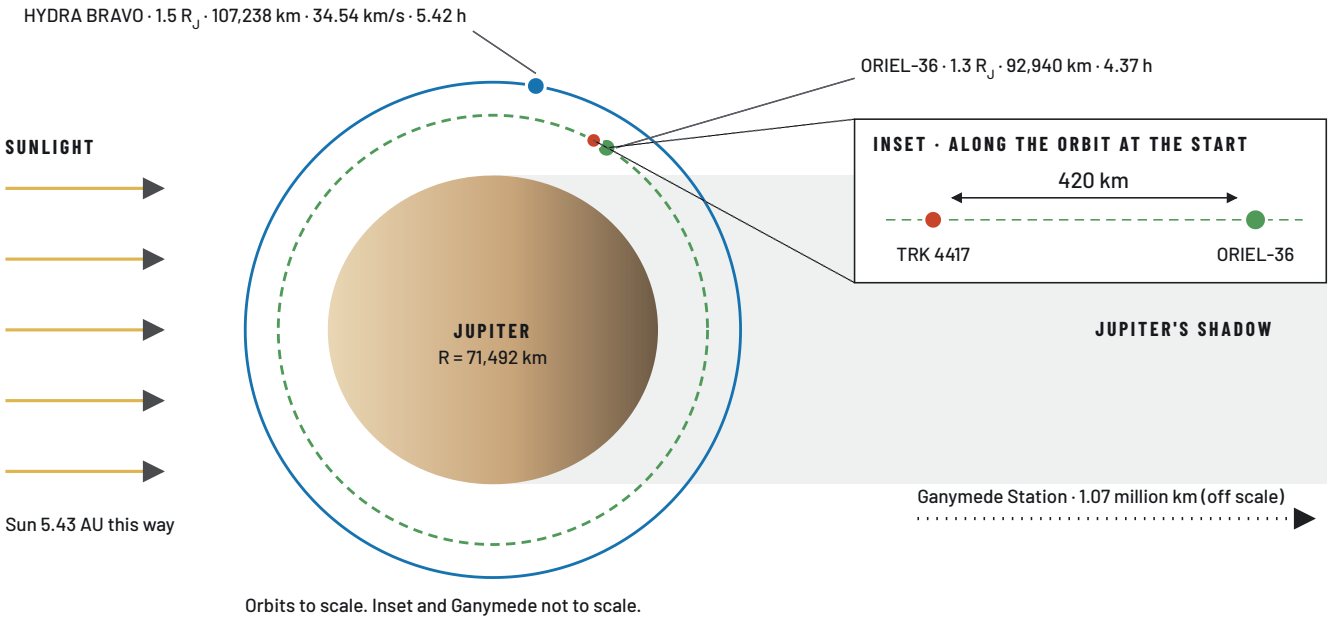
- `V-TEXT` The document was fetched and the passage read; the section, table or page is given.
- `V-RECORD` The work exists (publisher, DOI or official listing) but the passage was not read, because of a paywall or because it is a book. The equation used is the standard textbook form.
- `UNVERIFIED` Not confirmed. Treat it as unsourced.
- `DERIVED` Computed from verified inputs, with the method stated so it can be rerun.

Values chosen for play carry `DESIGN VALUE`. A design value is never presented as measured. Where possible, it is chosen inside a range that a source does support. The most important design values are the J/S needed to break a link (6 dB), the three hours of lost contact after which the inspector withdraws (only with a Blue ship within 25 km of it), the escalation costs, and the compressed timescale of the scenario.

The game runs in Godot 4.7 (GDScript, with 64-bit arithmetic wherever precision matters). Its physics is tested by an external suite (§10), which reads the game's own outputs. Python never reimplements the game in order to grade it.

2 FRAMES, TIME AND EPHEMERIDES

FIG. 2 THE SCENARIO, TO SCALE



HYDRA BRAVO and ORIEL-36 on their circular orbits around oblate Jupiter, with sunlight from the left and the shadow behind. Radii and speeds from §3 and §9; HYDRA's period is DERIVED as 2πr/v.

2.1 Reference frame

The simulation is centred on Jupiter. Its plane is Jupiter's equator, defined by the IAU 2015 pole (WGCCRE[7]), with x toward Jupiter's ascending node on the ICRF equator. The 3D view uses a floating origin at the camera's focus, so positions keep their precision at any distance. The game's clock is UTC. Leap seconds and TT − UTC are ignored, an error of 69.2 s, or 0.16° of Io's orbit.

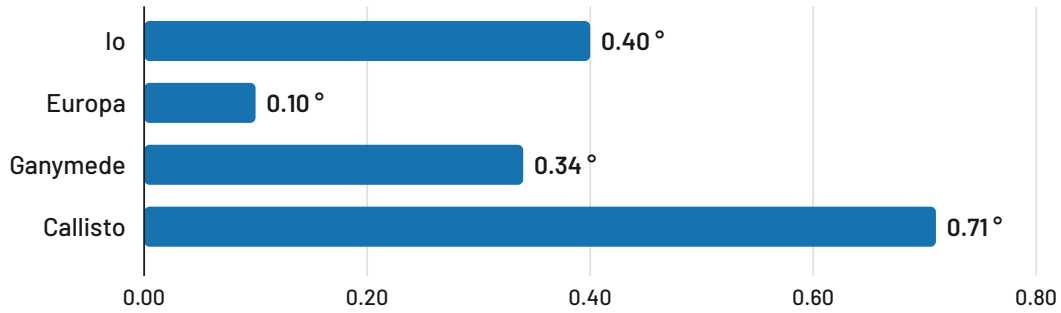
2.2 The Galilean moons

Each moon is a planar Kepler ellipse built from JPL's mean elements (JUP365). The mean longitude advances at the observed sidereal period (NASA's Jovian satellite fact sheet), and the mean anomaly at the period P that JPL tabulates, so periapsis precesses at the difference. JPL's legend calls P the sidereal period; the game uses it as the mean-anomaly period because, paired with the fact-sheet periods, that reproduces Horizons. An earlier build took the mean motion from Kepler's third law, $\sqrt{(\mu/a^3)}$. That ignores J2 and the Laplace resonance, and it put Io 114° from its true place at the campaign start. The current errors against JPL Horizons are:

MOON	PLANAR ERROR, CAMPAIGN	LONGITUDE, CAMPAIGN	LONGITUDE, 2036–2046	OUT OF PLANE (IGNORED)
Io	2,970 km	0.40°	0.43°	340 km
Europa	1,230 km	0.10°	0.24°	5,370 km
Ganymede	6,400 km	0.34°	0.38°	2,490 km
Callisto	23,200 km	0.71°	0.72°	9,900 km

Table 1. Moon positions against JPL Horizons. The Laplace angle $\lambda_I - 3\lambda_E + 2\lambda_G$ follows Horizons to 1.1°.

FIG. 3 **MOON LONGITUDE ERROR AT THE CAMPAIGN**



Against JPL Horizons, from Table 1. Callisto, the farthest, is worst at 0.71°.

2.3 The Sun and planets

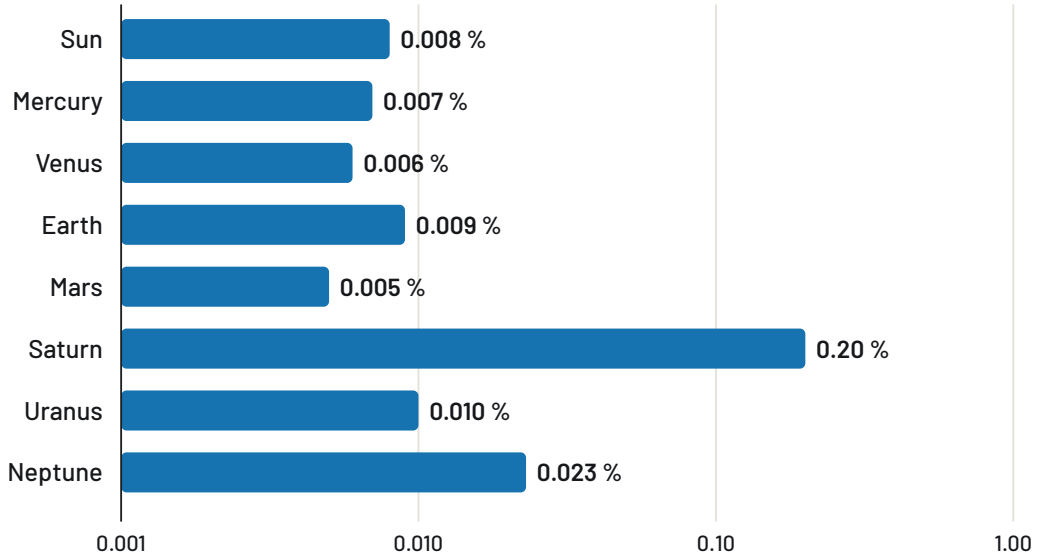
All eight planets come from JPL's approximate Keplerian elements (Standish & Williams[2]): Table 1 for 1800–2050, and Tables 2a and 2b otherwise. They are converted to Jupiter-centred positions in the game frame and cached on a cubic-Hermite grid every 6 h, which adds under 1 km of error. "Earth" in these tables is the Earth–Moon barycentre, 0.0003° from Earth as seen from 5 AU. GM values come from DE440[3]; radii from WGCCRE 2015[7] (Table 4, read through NAIF's pck00011 kernel); and the solar radius is the IAU 2015 nominal 695,700 km[8].

The verification suite checks every planet against Horizons over 2030–2050 and 2060–2199. As a spot check for this edition, the game's distances from Jupiter at the scenario's start (2041-10-12 16:10 UTC) were compared with Horizons vectors (centre 500@599, geometric):

BODY	GAME (AU)	HORIZONS (AU)	DIFFERENCE	ONE-WAY LIGHT TIME
Sun	5.4283	5.4279	0.008%	45.1 min
Mercury	5.7316	5.7312	0.007%	47.7 min
Venus	5.2125	5.2122	0.006%	43.4 min
Earth	6.3711	6.3705	0.009%	53.0 min
Mars	6.6496	6.6493	0.005%	55.3 min
Saturn	4.6212	4.6304	0.20%	38.4 min
Uranus	19.388	19.386	0.010%	2 h 41 min
Neptune	35.232	35.224	0.023%	4 h 53 min

Table 2. Planet distances from Jupiter at the scenario epoch. Saturn's 0.2% (1.4 million km) lies within the stated accuracy of the approximate elements. Saturn is unusually close because Jupiter and Saturn were in conjunction in late 2040.

FIG. 4 PLANET DISTANCES AGAINST HORIZONS



Difference in distance from Jupiter at 2041-10-12 16:10 UTC, from Table 2 (log scale). Every body is within 0.03% except Saturn.

3 GRAVITY AND TRAJECTORY DESIGN

3.1 Force model and integration

The acceleration on a craft is Jupiter's point mass, its J2 zonal term, and the four Galilean moons as third bodies (direct minus indirect term). The Sun is always on. Planets are added beyond 3 million km from Jupiter, where their tidal pull starts to matter. Inside that radius it is under 10^{-7} of Jupiter's (Montenbruck & Gill[20] §3.3, Cowell's method):

$$a = -\mu\, r/|r|^3 + a_{J2}(r) + \Sigma_k\, \mu_k\, [\, (r_k - r)/|r_k - r|^3 - r_k/|r_k|^3\,]$$

Jupiter's GM is 126,687,000 $km^3\ s^{-2}$ (NASA fact sheet[9]; 3.7×10^{-6} from JPL's JUP365). J2 is $14{,}696.514\times10^{-6}$ at R = 71,492 km (Juno, Folkner 2017 Table 2; 3.9×10^{-6} from Iess 2018 Table 1). Integration is RK4 with 60 s substeps near Jupiter. Beyond 3 million km the step adapts to 0.004 × the shortest free-fall time $\sqrt{(d^3/\mu)}$ to any body (Aarseth[21]). The coefficient is a DESIGN VALUE. Against DOP853 at rtol 10^{-13}, the integrator stays within 0.03 m over a day.

3.2 Two-body tools

Kepler propagation uses universal variables (Curtis[17] Alg. 3.3–3.4), with Danby's starter near e = 1 and Vallado's starter for long hyperbolic coasts (Vallado[18] Alg. 8). Lambert's problem is solved by Curtis Alg. 5.2. An earlier loss of precision in the Stumpff function c(z) as $z \to 4\pi^2$ made every transfer over 180° fail. The half-angle form $2\sin^2(\sqrt{z}/2)/z$ fixed it, and all 129 test transfers now agree with Izzo's solver (hapsira) to 1.9×10^{-12}.

3.3 Refining under full gravity

At 1.3 Jupiter radii, J2 bends a 3-hour coast by thousands of kilometres, so a two-body Lambert solution misses its target. Every planned burn is therefore refined by differential-correction shooting through the full-gravity coast (Vallado[18]; single shooting as set out in NASA/TP-20220014814 §5.2.1). Newton's method starts from a two-body Jacobian and updates it with Broyden's method[24]. HYDRA's transfer to ORIEL-36 flies at 3.570 km/s, against 3.869 km/s two-body, and arrives 0.012 km from its aim point. Circular orbits are initialised with $v^2 = \mu/r + 1.5\,\mu\,J2\,R^2/r^3$, so they don't oscillate.

3.4 The planner

The planner shows a porkchop plot: departure wait against time of flight, each cell a refined transfer coloured by the Δv to go and the Δv to match on arrival. For a target that circles in under 12 hours, waits run to 24 h in 1.5 h steps and flights from 1 to 6 h. Those grid steps are design values. The cheapest rendezvous with ORIEL-36 at the start is a wait of 18 h and a flight of 3 h: 1.90 km/s planned, 1.83 km/s flown, then 1.38 km/s to match.

3.5 Interplanetary flight

Where a campaign allows it, the planner plans between planets with patched conics (Curtis ch. 8; Vallado ch. 12). A heliocentric Lambert arc runs between the planets. The escape burn is fitted so its hyperbola leaves from the ship's actual position. Arrival is costed as capture into an orbit at 1.5 planet radii, a design value. The plan is then refined under full gravity, which is necessary because the Sun's tide during escape bends the departure by about 0.3 km/s. Spheres of influence follow Laplace, $r = a(m/M)^{2/5}$ (Curtis §8.4).

The Solar System window shows a minimum-energy (Hohmann) estimate for every body:

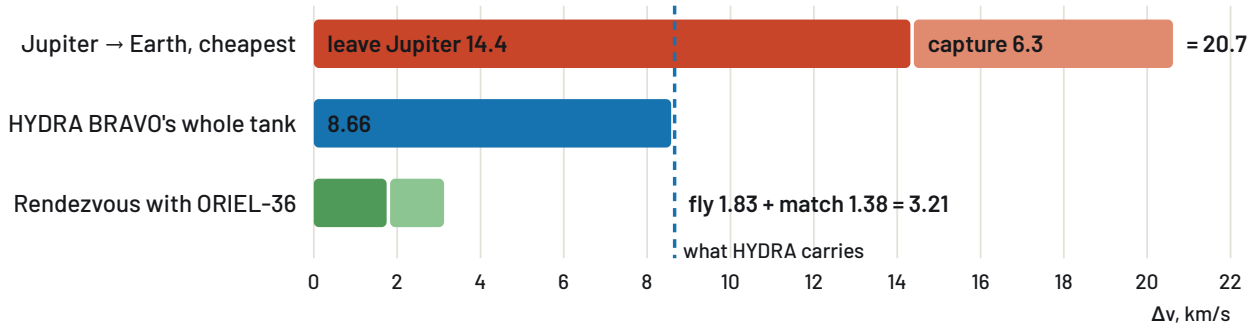
$$t = \pi \sqrt{(a_t^3/\mu_\odot)}, \quad a_t = (r_1 + r_2)/2$$

$$v_\infty = \sqrt{(\mu_\odot/r_1)}\ \left|\sqrt{(2r_2/(r_1+r_2))} - 1\right|$$

$$\Delta v_{leave} = \sqrt{(v_\infty^2 + 2\mu_J/r_s)} - v_s$$ (burn at the ship's radius r_s and speed v_s)

HYDRA orbits at 107,238 km from Jupiter's centre at 34.54 km/s, where the local escape speed is 48.6 km/s. Leaving Jupiter for Earth therefore costs about 14.4 km/s, plus 6.3 km/s to be captured at Earth, against HYDRA's 8.66 km/s. Interplanetary flight is switched off in the first scenario, and these figures show it would be out of reach anyway. That is real physics, not a game limit. From Jupiter's orbit, the cheapest flight takes 3.2 years to Mars and 2.9 years to Earth, and a fall into the Sun takes 2.2 years.

FIG. 5 **WHY HYDRA STAYS AT JUPITER**



Minimum-energy Δv from §3.4–3.5. Leaving Jupiter from HYDRA's orbit costs more than its whole tank; the scenario's rendezvous costs about a third of it.

4 RADIO: ANTENNAS, LINKS AND DETECTION

4.1 Antenna gain and patterns

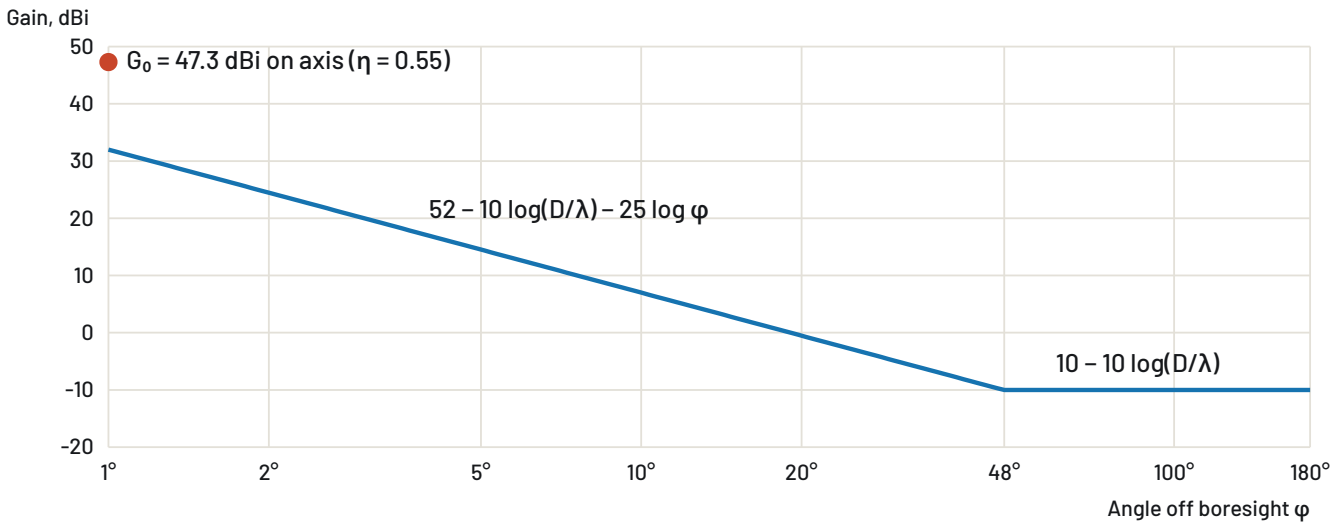
The peak gain of a dish of diameter D at wavelength λ is $G_0 = \eta(\pi D/\lambda)^2$, with $\eta = 0.55$ throughout. That is 1.24 dB conservative against the DSN 34 m antenna at X band (68.24 dBi at 8425 MHz, DSN 810-005 Module 104 Rev. O Table 8[39]). The beamwidth is about 70λ/D degrees (NAWCWD TP 8347 §3-1[54]; ITU-R SA.509-3 §1.3[33]) V-TEXT.

Off-axis gain follows the ITU reference envelopes: SA.509-3 for large dishes ($D/\lambda \geq 100$), and S.465-5 Note 4 for small ones[34]:

```
G(ϕ) = 52 − 10 log(D/λ) − 25 log ϕ   dBi,   100λ/D ≤ ϕ < 48°
G(ϕ) = 10 − 10 log(D/λ)              dBi,   48° ≤ ϕ ≤ 180°
```

These are envelopes of sidelobe peaks, used in interference coordination, so they are conservative for a well-designed antenna. Note 2 of S.465 warns against using them below $D/\lambda = 50$, and every Red dish in the game is that small. They are still the most credible public pattern. Every emitter points: Red's telemetry and command antennas at their handler, civil probes at Earth, and the ore hauler at Ganymede. Warships and buoys listen through four-face arrays that form beams in all directions (after the TDRS S-band multiple-access array, NASA TM-105422). Their gain is the face's peak plus 10 log cos(scan). Ganymede Station's 12 m dish has to be pointed: an assist cues it to the most relevant track, and the player can override it.

FIG. 6 REFERENCE ANTENNA ENVELOPE



ITU-R SA.509-3 sidelobe envelope drawn for an illustrative dish 100 wavelengths across. It is an envelope of sidelobe peaks, so it is conservative for a well-designed antenna.

4.2 Link budget and noise

$$P_r = P_t\, G_t(\theta_e)\, G_r(\theta_r)\, (\lambda / 4\pi R)^2 \qquad \text{(Friis 1946)}$$

$$N = k\, T_s\, B$$

Free-space loss is the exact $20 \log(4\pi R/\lambda)$ of ITU-R P.525-4 eq. (3). The practical eq. (4) rounds its constant to 32.4 dB (exactly 32.448 dB), so the game differs from eq. (4) by 0.048 dB.

4.3 Detection of unknown bursts

Listeners don't know a burst's waveform, so they detect it with an energy (radiometer) detector (Urkowitz 1967[51]). The integrated SNR grows as $\sqrt{(B\tau)}$. The chance of detection is:

$$P_d = Q\left(\left(Q^{-1}(P_{fa}) - d\right) / \sqrt{(1 + 2s)} \right), \qquad P_{fa} = 10^{-6}$$

The false-alarm rate is a **DESIGN VALUE**. This expression matches the exact chi-square distributions to 0.01 for $n \geq 10^5$ samples, and even odds come at 6.8 dB. Spreading a signal over a wider band costs an eavesdropper's energy detector $\sqrt{(B_{old}/B_{new})}$ in SNR at fixed power (the radiometer equation; Urkowitz 1967[51]). ITU-R SM.1055[38] §1 notes that spread signals are noise-like and not readily detected.

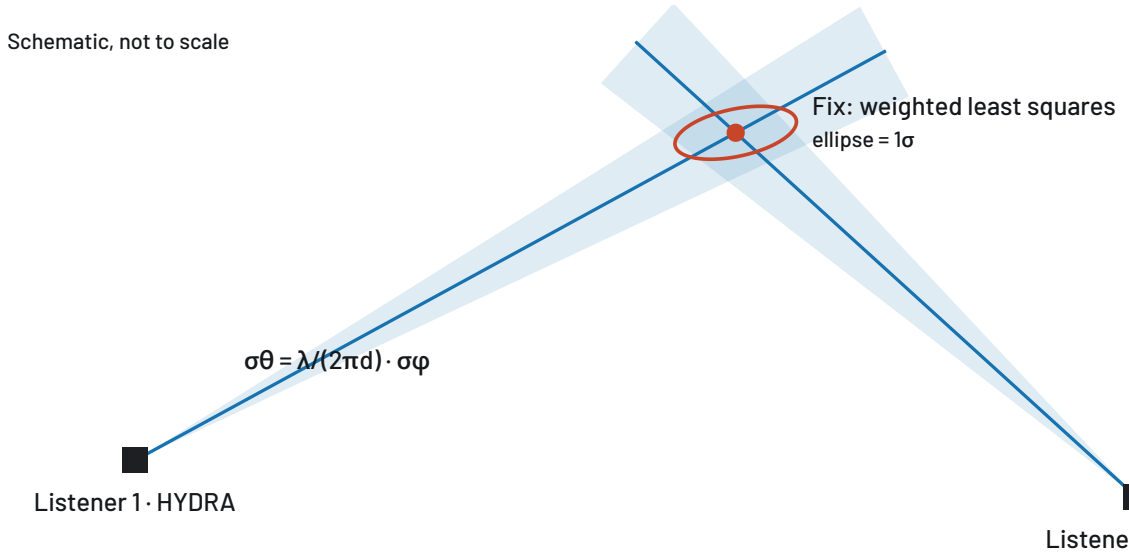
4.4 Direction finding and identification

A two-element interferometer measures a bearing with

$$\sigma_\theta = \lambda/(2\pi d)\cdot\sigma_\phi, \qquad \sigma_\phi^2 = (1+2s)/(2B\tau\, s^2)$$

This is the cross-correlation phase error for a noise-like signal. It follows from the Knapp & Carter (1976) Cramér–Rao bound in terms of coherence, $\gamma^2 = s^2/(1+s)^2$, in the narrowband limit; it is checked by a Monte Carlo of a two-element correlator to 3%. A 20 µrad floor is a DESIGN VALUE. Bearings from separated listeners are crossed by weighted least squares. Identification compares a recording's modulation and symbol rate against a threat library above a minimum SNR, a design-value gate. Specific-emitter identification is real: it recognises a device by imperfections in its analog circuitry (Danev et al.[78]). Once a signal is identified, the coherent turnaround ratio $f_{up} = f_{down} \times 749/880$ (CCSDS 401 Rec. 2.6.2; DSN 810-005[39]) gives its command uplink.

FIG. 7 **DIRECTION FINDING**



Two listeners each take a bearing with its own error wedge. The fix is their weighted least-squares crossing; the farther apart the listeners, the tighter the ellipse. Schematic.

5 ELECTROMAGNETIC ATTACK

5.1 The jamming equation

Jamming a command uplink means being louder, at the victim's receiver, than its handler. Both paths are one-way, so the equation is Friis applied twice. This is the communications form of NAWCWD TP 8347 §4-9 V-TEXT, and

Adamy's EW 101[75] gives it in dB form:

$$J = P_j + G_j(\text{aim error}) + G_v(\theta_j) - L_j$$

$$S = P_u + G_u + G_v(0) - L_s$$

$$J/S = J - S - 10\log(B_j/B_v) \quad \text{dB}$$

$G_v(\theta_j)$ is the victim's gain toward the jammer, at angle θ_j off its line to the handler. From broadside, the victim's discrimination is −44.8 dB. Near the handler's line it falls to −13 dB, so geometry matters. The aim term means a loose track costs beam-pointing gain. HYDRA's jammer is 1 kW from four combined 250 W TWTAs into a 1.5 m dish at 7.18 GHz, 68.5 dBW EIRP. That is a plausible descendant of the Counter Communications System. Real CCS and Meadowlands power figures are not public, so this is an extrapolation DESIGN VALUE.

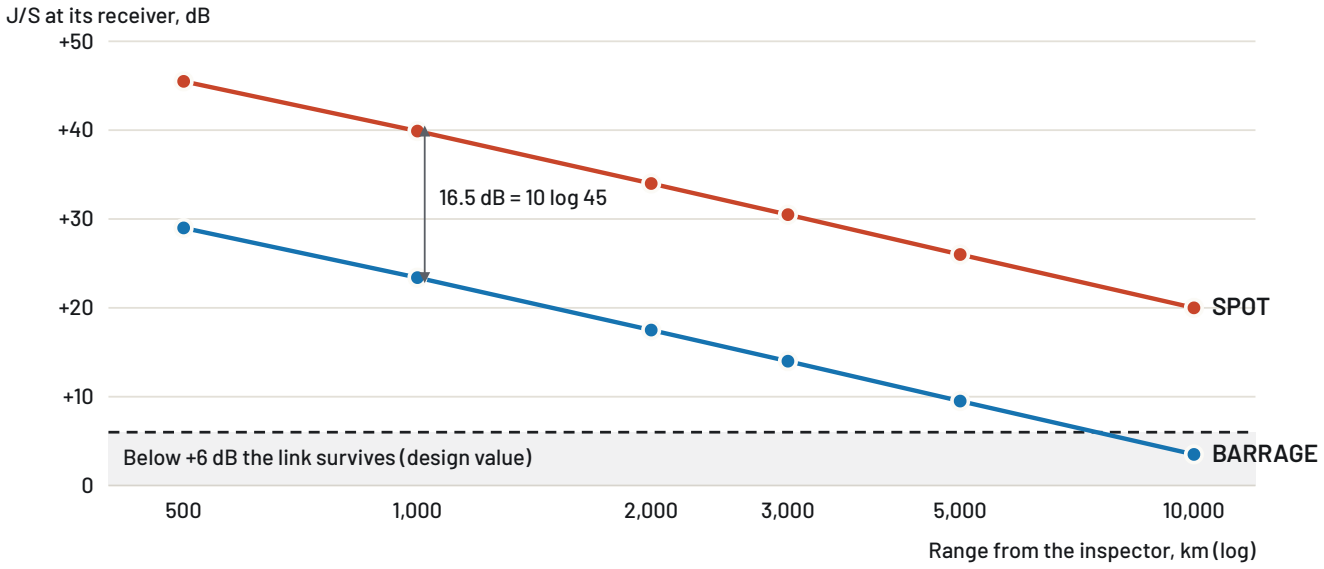
5.2 Spot and barrage

Before the signal is identified, the jammer must spread its power across the whole deep-space X uplink allocation, 7145–7190 MHz (DSN 810-005 Module 201 Rev. F V-TEXT). That costs 10 log(45) = 16.53 dB against a 1 MHz channel. An earlier build used 90 MHz, which included the near-Earth 7190–7235 MHz allocation. Spot jamming of the identified channel removes the penalty. Measured from beside the inspector, barrage gives:

RANGE	500 KM	1,000	2,000	3,000	5,000	10,000
Barrage J/S	+29.0 dB	+23.4	+17.5	+14.0	+9.5	+3.5
Spot J/S	+45.5 dB	+39.9	+34.0	+30.5	+26.0	+20.0

Table 3. J/S against the inspector's command uplink. Spot is barrage + 16.5 dB.

FIG. 8 JAMMING THE INSPECTOR'S COMMAND UPLINK



J/S against range from Table 3. Spot jamming, possible only once the signal is identified, gains 16.5 dB over barrage. The +6 dB threshold is a DESIGN VALUE. Hover a point for its value.

How much J/S breaks a link has no universal number. NAWCWD §4-7 (p. 4-7.3) says plainly that effectiveness "is not a direct mathematical function of J/S" V-TEXT. The game requires +6 dB, a DESIGN VALUE inside the 0–10 dB band typical for FEC-coded BPSK.

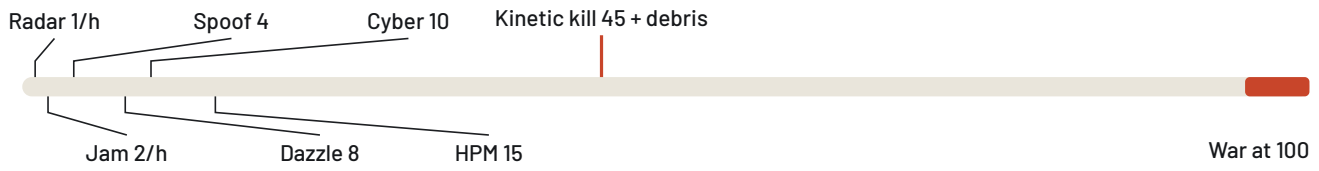
5.3 Other effects

EFFECT	MODEL	SOURCE / STATUS
Spoof (FAKE)	False ranging tones capture the target's ranging loop at J/S ≥ 0 dB, inducing a drift	Loop capture at 0 dB and a drift of 900 km/h are design values
Dazzle	Spot radius $R\sqrt{((1.22\lambda/D)^2 + j^2)}$ at 1.06 µm, 2 m aperture, 0.2 µrad jitter; 10 W/m² at the aperture	Airy disc $1.22\lambda/D$ (Hecht); jitter added in quadrature (engineering approximation); 10 W/m² is an engineering estimate allowing for an OD-6 filter
HPM (PULSE)	Far field $E = \sqrt{(30PG)}/R$ from a 1 GW-peak X-band source; front-door damage at 100 V/m	$E = \sqrt{(30PG)}/R$ (Balanis; ITU-R P.525-4 eq. 1); 100 V/m is a DESIGN VALUE (engineering estimate); GW-class backward-wave oscillators (Benford et al. 2016)
Cyber (TAKE)	Takeover after recording enough command traffic	Abstracted. Real command links are encrypted, and this is stated as fiction
Kinetic kill	$P_k = \text{rel}\cdot(1 - e^{-r^2/2\sigma^2})$, from divert reach against track uncertainty	Rayleigh. Breakup by the NASA standard breakup model (below). The 1,200 catalogued pieces are a DESIGN VALUE set below FY-1C (2007; 3,442 catalogued[138]) and Cosmos 1408 (2021)

A kill breaks the wreck up by the NASA standard breakup model (Johnson et al. 2001[131], as given in Frey & Colombo 2021 App. B[135]). An impact is catastrophic above 40 J/g, and $N(\geq L_c) = 0.1\ M^{0.75}\ L_c^{-1.71}$ fragments are drawn. Each fragment's area-to-mass ratio $\chi = \log_{10}(A/M)$ follows App. B: the bimodal spacecraft law above 11 cm, the small-object law below 8 cm, and a linear blend in $\log L_c$ between 8 and 11 cm. Its speed follows $\log_{10} \Delta v \sim N(0.9\chi + 2.9, 0.4)$, Δv in m/s.

Escalation costs are design values: jamming 2 per hour, radar 1 per hour, spoofing 4, dazzling 8, cyber 10, HPM 15, and a kinetic kill 45. Doctrine provides no escalation ladder (§8).

FIG. 9 THE ESCALATION LADDER



Jamming alone, all day: 2 × 24 = 48 a day. The scenario lasts three days. All costs are DESIGN VALUES.

The costs from §5.3. Doctrine provides no ladder (§8): its ordering, reversible and deniable acts cheap, debris ruinous, is the game's.

6 HEAT, INFRARED AND VISIBLE SIGNATURES

6.1 Energy balance

Each craft has a skin and, if it carries a reactor, radiators. Skin temperature follows a single-node balance with heat capacity (Gilmore[103]):

$$C\,dT/dt = P_{leak} + \alpha\,S\,f\,A/4 - \varepsilon\,\sigma\,T^4\,A, \qquad C = 897\ \mathrm{J\,kg^{-1}\,K^{-1}} \times 0.25 \times m_{dry}$$

S = 50.26 W m^{-2} at Jupiter (NASA fact sheet V-TEXT), and f is the sunlit share (§6.4). The heat capacity is aluminium's at 298 K, 24.21 J mol^{-1} K^{-1} or 897 J kg^{-1} K^{-1} (NIST-JANAF, NIST Chemistry WebBook[105]); Incropera[104] Table A.1 gives 903 J kg^{-1} K^{-1} at 300 K, 0.7% more. The the 25% skin fraction is a DESIGN VALUE. HYDRA's thermal time constant is about 22.5 h, so a pass through Jupiter's shadow cools it only from 190 K to about 187 K. Absorptance and emittance come from NASA RP-1121 (Henninger 1984[95]).

Reactor ships use the 100 kWe Brayton plant of NASA Glenn's JIMO power-system study, at 20.2% efficiency (Mason 2003[96]). That rejects 395 kW, all of it through two-sided Lambertian radiators (JIMO's main radiator carried 364 kWt; the rest left by other routes), $I(\theta) = \varepsilon\sigma T^4 A \cos\theta/\pi$. WARDEN's 85 m^2 per face runs at 462 K, within 1.1% of JIMO's design point (364 kWt from 170 m^2, Siamidis 2005[98]). Radiators lie flat to the plane of play, so every observer in the plane sees only their edge.

6.2 Going dark

A ship can stop radiating for a while by storing its heat. WARDEN evaporates and vents 2 t of water (2.5 MJ/kg, the latent heat near 0 °C), which carries 395 kW for 3.5 h and is then gone. VESPER melts 500 kg of ice (333 kJ/kg, Leimkuehler 2011[102]) for 15.4 h and refreezes it afterwards.

6.3 In-band infrared detection

Telescopes work in NEO Surveyor's NC2 band, 6–10 µm (Mainzer et al. 2023[115] V-TEXT). The in-band share of each body's emission comes from the Widger & Woodall series, which matches astropy to 6×10^{-11}. The 0.5 m LWIR's 1σ noise-equivalent irradiance is 7×10^{-18} W m^{-2}. That lies inside the 0.4–1.1×10^{-17} range derived from NEO Surveyor's S/N-5 sensitivity. A look detects with

$$S/N = I / R^2 / (NEI \times background), \qquad P_d = \Phi(S/N - 5)$$

Against Jupiter's disc, the background raises the NEI ×50, a DESIGN VALUE inside the ×26–92 derived for a background-limited sensor. Within 3 Jupiter radii off the disc, stray light raises it ×5, also a design value. False

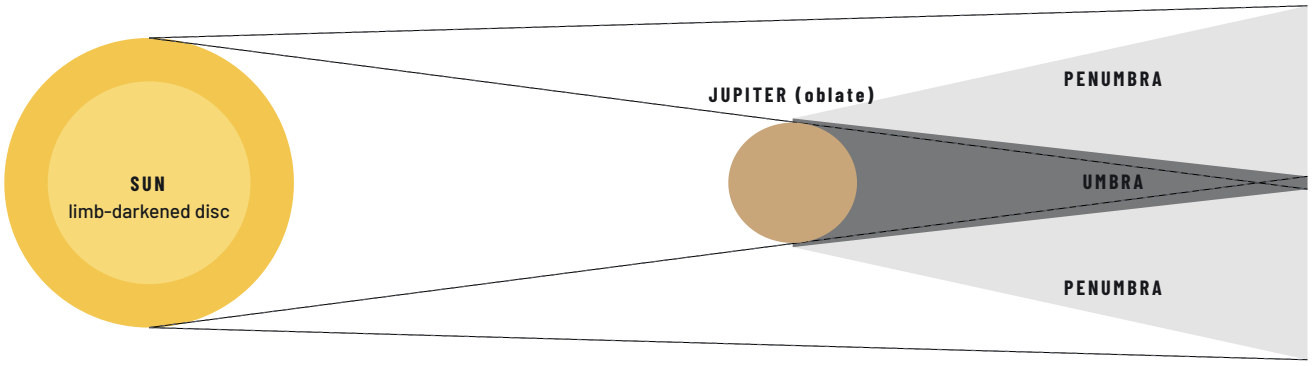
alarms are Poisson, with a mean of 0.1 per sensor per look (design value), and only lines from two places at once start a track.

6.4 Shadows and reflected light

The Sun is a limb-darkened disc, $I(\mu) = 1 - u(1 - \mu)$ with $u = 0.6$ (Allen's[106] §14.7), sampled by 397 rays in 12 rings. The sunlit share is the weighted fraction not blocked by oblate Jupiter (71,492 × 66,854 km), the four moons, or, far out, a planet. This gives both umbra and penumbra (Montenbruck & Gill §3.4.2), within 0.019 of a dense numerical integration.

Sunlit hulls are also seen in visible light. The Sun's $M_V = 4.81$ (Willmer 2018) is scaled to Jupiter's irradiance, and each hull is a Lambert sphere with phase law $\Phi(\alpha) = (\sin\alpha + (\pi - \alpha)\cos\alpha)/\pi$. It returns a share $p \cdot \Phi(\alpha) \cdot r^2/R^2$ of the sunlight at its distance R, with p the geometric albedo and $r^2 = A/4\pi$ for a hull of surface area A (Madhusudhan & Burrows 2012 eq. 3[122]). An earlier build used the cross-section πr^2 in place of r^2, which made every hull π times (1.24 mag) too bright and every visible range $\sqrt{\pi} = 1.77$ times too long. A 1 m GEODSS-class telescope reaches magnitude 22 (Stokes et al. 1996[119]; the free 1998 Lincoln Laboratory Journal account of the same tests gives magnitude 22 at S/N 4 in under 100 s). The game asks for S/N 5 at magnitude 22, a DESIGN VALUE. Visible telescopes look only where a track already exists, because reaching magnitude 22 takes about 100 s staring at one field.

FIG. 10 **UMBRA AND PENUMBRA**



Schematic, not to scale. 397 rays in 12 rings sample the disc; the share not blocked is the sunlit fraction f.

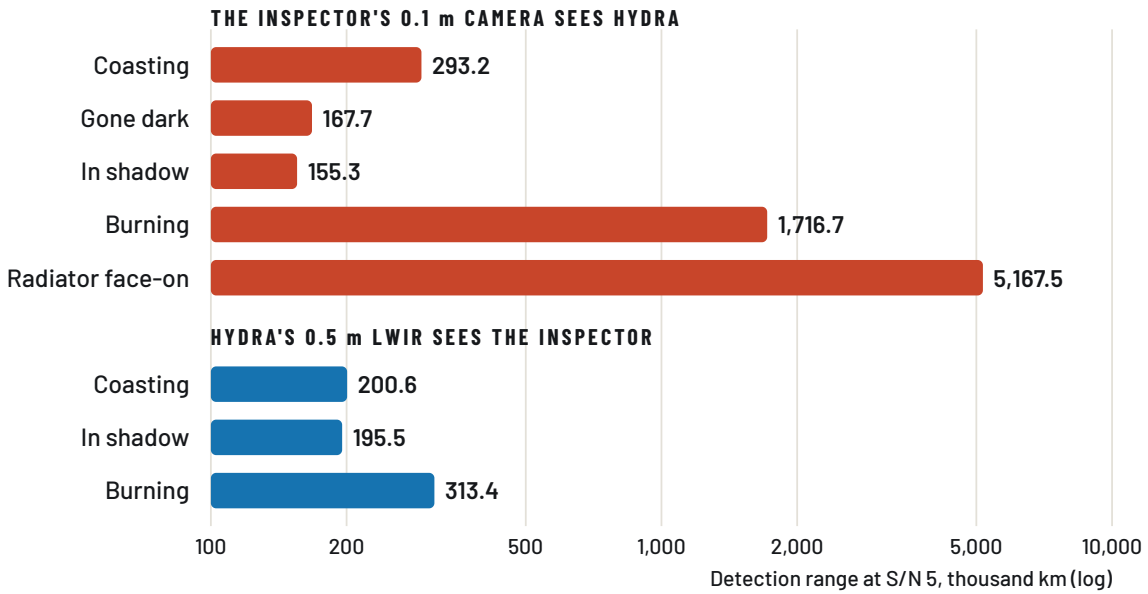
Because the Sun is a disc, Jupiter casts a full shadow inside a soft-edged one (§6.4).

6.5 What the scenario's sensors see

OBSERVER → TARGET	COASTING	DARK	RADIATOR FACE-ON	BURNING	IN SHADOW
HYDRA's 0.5 m LWIR → the inspector	200.6	200.6	200.6	313.4	195.5
The inspector's 0.1 m camera → HYDRA	293.2	167.7	5,167.5	1,716.7	155.3
The inspector's camera → ORIEL-36	181.8				

Table 4. IR detection ranges at S/N 5, thousand km. HYDRA starts out of the inspector's sight, and is seen far sooner coasting than dark.

FIG. 11 **WHO SEES WHOM, AND HOW FAR**



Infrared detection ranges from Table 4 (log scale). Going dark cuts the range at which the inspector sees HYDRA by 43%; a radiator turned toward it makes it about eighteen times longer.

With a 1 m telescope, the visible detection ranges after the corrected Lambert-sphere formula (§6.4) are:

HULL	VISIBLE RANGE, 1 M (THOUSAND KM)
GADFLY	384
CROW	470
WREN ES/DF buoy	271
OWP-62 WARDEN	1,213
PFS	2,093
HMH-5 heavy hauler	3,309
GMC-2 ore carrier	632
LANTERN relay	509
OX ore tug	770
PROSPECTOR probe	272

Table 4b. Visible detection ranges with a 1 m aperture, thousand km, with the $r^2 = A/4\pi$ correction (each is 1/1.77 of the earlier build's).

7 TRACKING

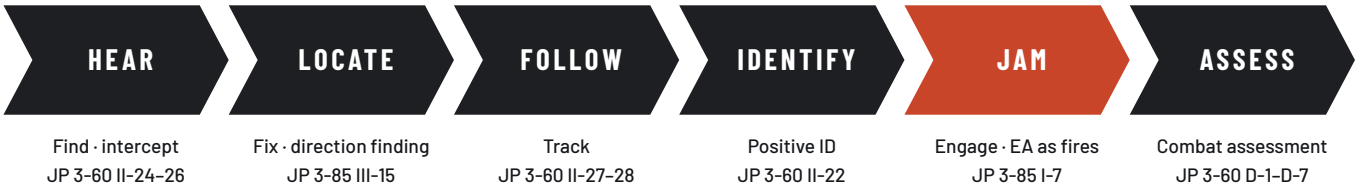
Every contact is a track: an estimate with a covariance, not ground truth. Bearings update it through an unscented Kalman filter (Julier sigma points, $\kappa = 0$, circular mean for angles), which matches filterpy to 3×10^{-12}. Prediction uses central-difference state transition matrices and white-noise acceleration (Bar-Shalom et al. 2001). Two cases needed special handling.

- **Close in.** With HYDRA a few kilometres from the inspector, sigma points can straddle the sensor. Within a tenth of the range, a bearing therefore enters as a pseudo-linear measurement (Lindgren & Gong 1978; Aidala & Nardone 1982). Between craft in nearly the same orbit, range is unobservable from angles alone without a manoeuvre (Woffinden & Geller 2009). So without a recent radar range, the track keeps at least 0.5 × range of 1σ along the line of sight (design value). In 42 looks from alongside to departure, no estimate fell outside 3σ.
- **Unseen burns.** A bearing whose normalised innovation passes the $\chi^2(1)$ 99.9% point (10.83) is taken as a burn nobody saw, and widens the track (Bar-Shalom ch. 11). The 50 m/s and 10 min used are design values.

A track nobody has detected for 24 h, or whose 1σ exceeds 100,000 km, is dropped (after Blackman & Popoli 1999). Both thresholds are design values. Contacts are drawn after MIL-STD-2525D, with a dashed frame for a last known position.

8 DOCTRINE AND THE KILL CHAIN

FIG. 12 THE GAME'S VERBS AS A KILL CHAIN



Each step mapped to its doctrinal term and citation (table below).

The game's verbs follow US joint doctrine, read from the published texts: JP 3-85 *Joint Electromagnetic Spectrum Operations* (2020)[139], JP 3-60 *Joint Targeting* (2018)[141], AFDP 3-60 (2026)[142] and the USSF *Space Warfighting* framework (2025)[143].

GAME STEP	DOCTRINAL TERM	CITATION
Hear	F2T2EA *Find*; electromagnetic support search and intercept	JP 3-60 II-24–26; JP 3-85 I-6
Locate	*Fix*; direction finding, precision geolocation	JP 3-60 II-26–27; JP 3-85 III-15, B-4
Follow	*Track*	JP 3-60 II-27–28
Identify	Positive identification by EW support systems or visual recognition (doctrine acquires it during *Fix*)	JP 3-60 II-22
Jam	*Engage*; electromagnetic attack as a form of fires	JP 3-85 I-7, I-13, GL-6, GL-7
Assess	Combat assessment, with confidence graded confirmed, probable or possible	JP 3-60 D-1–D-7

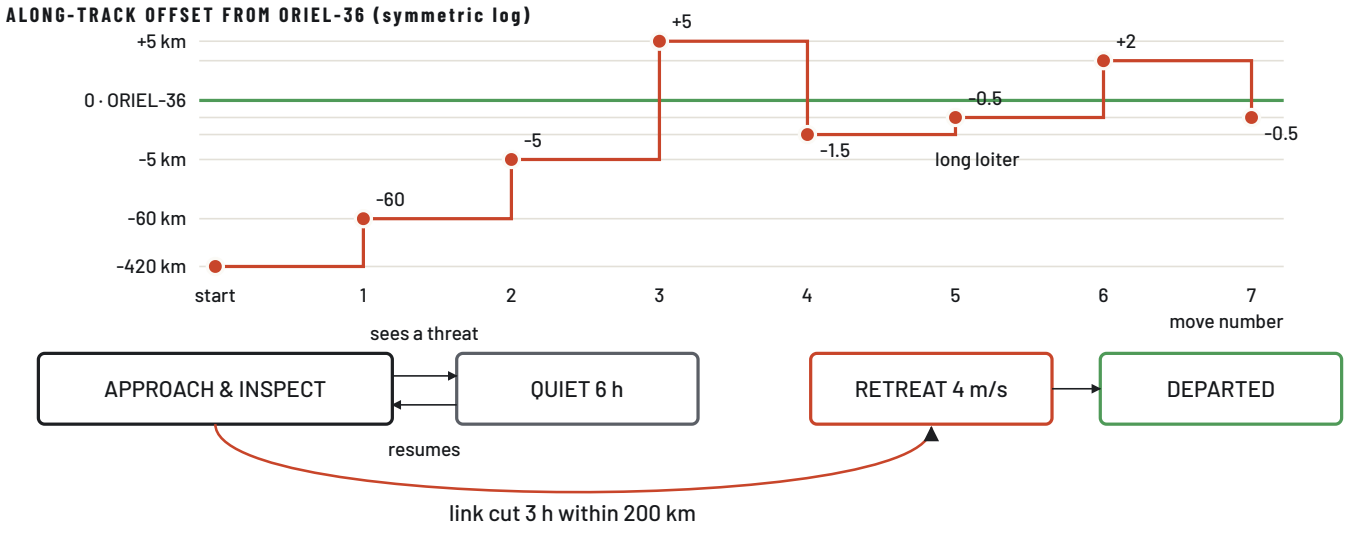
Under JP 3-85, a stealth ship going dark is practising electromagnetic protection: "multispectral low-observability" is one of its listed examples. Doctrine says nothing about TDOA/FDOA, look-through, own-platform jammer isolation, or an escalation ladder. JP 3-85, JP 3-60 and AFDP 3-60 never use the words "spoofing" or "reversible": their terms are intrusion and deception, and temporary or nonlethal effects. (The USSF framework does speak of reversible and nonreversible actions, pp. 5 and 9.) The escalation meter is therefore entirely a design value. Its ordering (reversible and deniable acts cheap, debris-generating acts ruinous) follows doctrine's escalation-of-force guidance qualitatively.

9 THE SCENARIO AND ITS REAL-WORLD BASIS

THE SHADOWING is modelled on a real uncooperative rendezvous and proximity operation against a commercial X-band radar-imaging satellite in October 2026. It was reported from LeoLabs tracking data by Breaking Defense and The War Zone on 19 October 2026[158,159]. The events are moved to Jupiter and the craft are fictional. As reported, the real sequence was: coplanar alignment; closing from long range; station-keeping within 5 km, then 1–2 km; a closest pass reported at 500 m (another report gives 26 m); moves ahead of and behind the target; and a controlled departure into a larger orbit. The inspector's seven moves follow that sequence: −60, −5, +5, −1.5, −0.5 (the long loiter), +2 and −0.5 km along-track.

- **Orbit.** 1.3 R_J (92,940 km), a 4.37 h period. This is inside Jupiter's radiation belts, so every craft is assumed rad-hardened (Juno's titanium vault cuts its dose about 800 times[161]). No dose is modelled.
- **The inspector's radio.** An X-band (8.436 GHz) BPSK burst at 256 kbaud, 10 W into a 0.4 m dish aimed at its handler, 90 s long, after each move. These are design values within ordinary CCSDS practice. Seeing a threat, it goes quiet for 6 hours.
- **Losing its link.** Cut off from its handler for 3 hours within 200 km of its target, and with a Blue ship within 25 km of it (a design value: one MATCH drifts about 20 km in 3 h), it backs away prograde at 4 m/s and departs. Cut off but alone, it holds its slot and waits. Taking its controls needs the jammer holding its owner off first. This is grounded in practice, not in a published rule: a Progress approach to the ISS was aborted on loss of its telemetry link in 2010[162], and ESA's ATV carried RETREAT and ESCAPE modes for the same reason. The threshold and the burn are DESIGN VALUES.
- **Timescale.** The real event took weeks. The game compresses it to about 57 hours over three days DESIGN VALUE.

FIG. 13 HOW THE INSPECTOR BEHAVES



Top: the seven along-track station points from §9, on a symmetric log scale so the last few hundred metres are visible. Bottom: what makes it go quiet or leave. Thresholds are DESIGN VALUES.

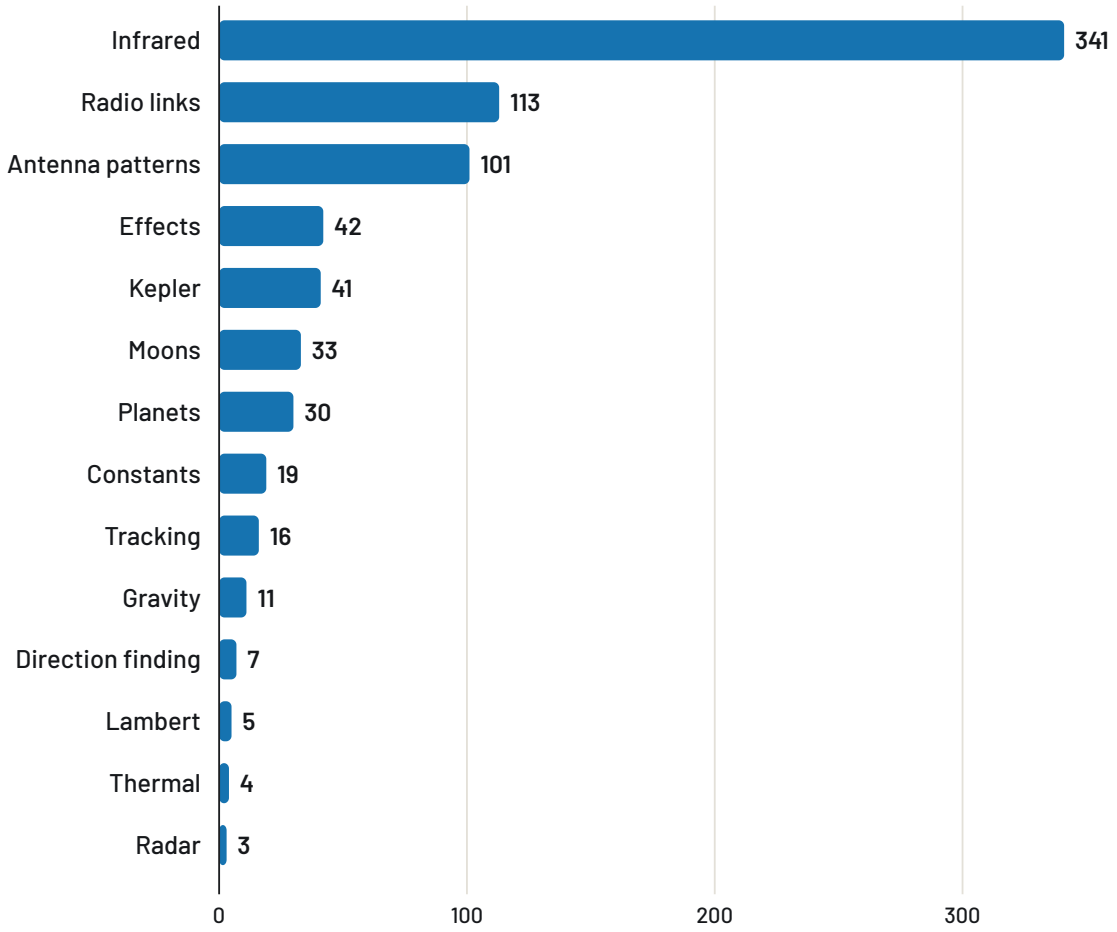
10 VERIFICATION

A Godot script runs every test case through the game's own functions and writes the results out. A Python suite (pytest) then compares them with independent references, each comparison carrying its own tolerance. JPL data is cached, so the suite runs offline. On 25 October 2026 it made 762 comparisons across 63 tests; all 762 fall within tolerance, and 4 further rows report known simplifications.

MODEL	REFERENCE	TOLERANCE	WORST RESULT
Kepler propagation (122 cases, e to 0.999 and hyperbolic)	hapsira farnocchia/vallado; DOP853	1e-8	2.3e-10
Lambert (129 transfers)	hapsira izzo	1e-6	1.9e-12
Worked examples	Curtis Ex. 3.7, 5.2; Vallado Ex. 2-4	½ last digit	pass
Force model (J2, third bodies)	hapsira perturbations; gradient of the J2 potential	1e-9	9e-11
RK4 integration, 1 day	DOP853, rtol 1e-13	5 m	0.03 m
J2 secular rates	Vallado eqs. 9-41/42	1%	0.40%
Moons and planets	JPL Horizons	per body	Table 1
Antenna patterns (468 cases)	ITU-R SA.509-3, S.465-5, coded from the text	1e-9 dB	1.4e-14
Free-space loss	ITU-R P.525-4 eq. (4), 32.4 dB constant	0.05 dB	0.048 dB
Radiometer gain √(Bτ)	Monte Carlo energy detector, 2,000 trials	6%	1.5%
Interferometer DF σ	Monte Carlo two-element correlator	8%	3%
Radar SNR and monopulse angle	Skolnik; Barton	1e-9; 0.2%	6e-15; 0.14%
Jammer J/S and reach	Adamy, independent dB form; brentq	1e-9 dB	3e-14
In-band Planck share	astropy BlackBody, 60–3,000 K	1e-6	6e-11
Radiators against JIMO	Siamidis 2005 design point	2%	1.1%
Shadow fraction	Dense numpy disc integration	0.02	0.019
Skin heat capacity	scipy solve_ivp	—	0.0002 K
Unscented update	filterpy UKF	1e-6	3e-12
Rocket equation	Tsiolkovsky, every ship	0.006 km/s	0.0045 km/s

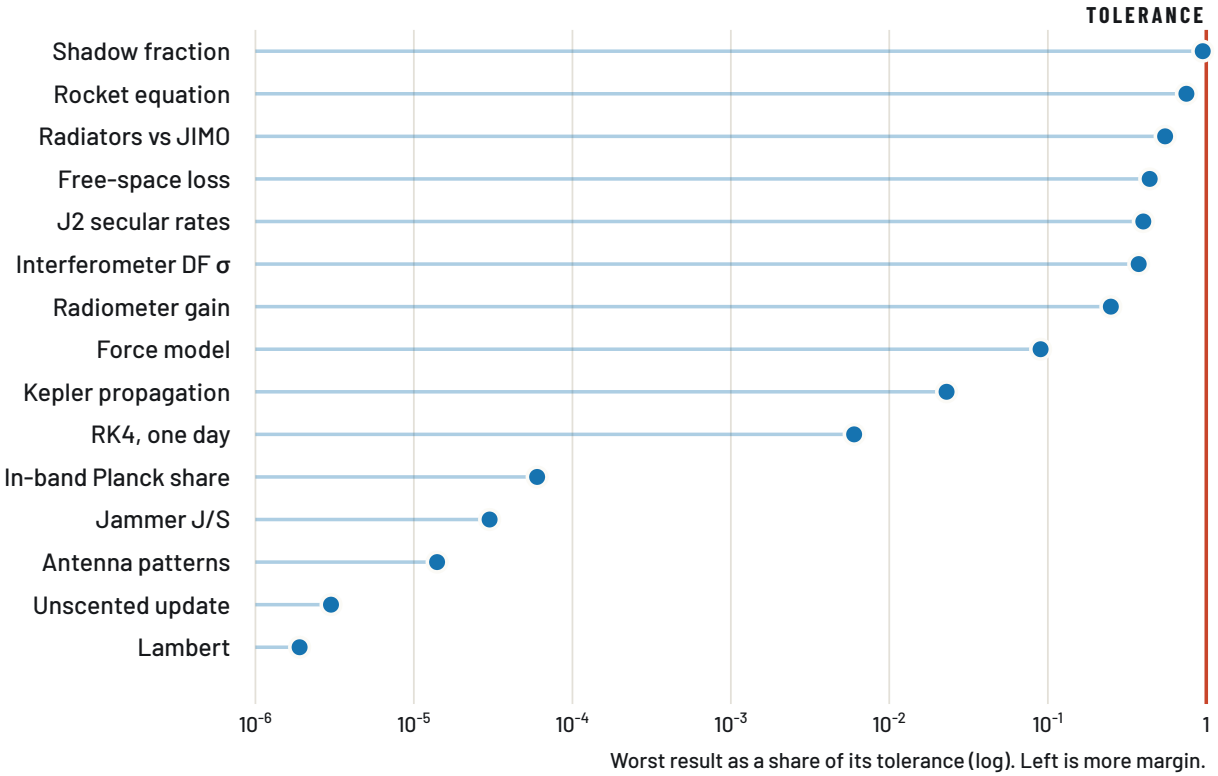
Table 5. Selected comparisons. The full table has one row per comparison.

FIG. 14 WHERE THE 766 ROWS GO



Rows per domain on 25 October 2026: 762 within tolerance, and 4 that report known simplifications.

FIG. 15 **HOW MUCH MARGIN EACH CHECK HAS**



Worst result divided by its tolerance, for the numeric rows of Table 5. Every dot sits left of the red line; most sit orders of magnitude inside it.

The suite has also found and fixed real errors. Moon longitudes were up to 114° off. DF σ was three times too large, because it was fed the integrated SNR. The barrage band was twice as wide as the real allocation. Long-way Lambert transfers failed. Kepler's equation lost accuracy near e = 1. Each fix and its effect on play is recorded in the project's verification log.

11 LIMITS, SIMPLIFICATIONS AND OPEN QUESTIONS

Not public, so extrapolated

- Jammer, uplink and handler EIRPs. Counter Communications System and Meadowlands power is not published.
- Weapon-class laser wall-plug efficiency. The game's 25% is conservative against industrial fibre lasers.
- Time-transfer accuracy between platforms at Jupiter, where there is no GPS.

Not verified

- Mid-infrared zodiacal background at 5.2 AU, and Jupiter's band-resolved brightness (a grey body at 125.6 K is used).
- HawkEye 360 geolocation accuracy; the Albersheim validity limits; specific-emitter identification accuracy; look-through duty cycles.
- The 2024 edition of JP 3-60 (the 2018 edition was read), and JP 5-0's definitions of MOE and MOP.
- MIL-STD-2525D was read through a reproduction of its tables, not the original PDF.

Simplifications

- Moons are planar mean-element ellipses with no mutual perturbations (≤ 23,000 km and ≤ 0.72°).
- Radiators lie flat to the plane, and the effect of jamming is binary (link cut or not). Doctrine describes a continuum from temporary to permanent and from partial to complete.
- The umpire's view: some kill-chain indicators still read ground truth rather than Blue's assessed belief.
- Red doesn't yet adapt by frequency hopping, nulling or switching to optical links. All of these are real (ITU-R SM.1055; Cummings 1992; NASA DSOC).
- No radiation dose is modelled.

---

12 XGEO CORE AS AN OPEN-SOURCE TOOL

The models in this edition make up **xGEO Core**, the engine under the game. **xGEO Live**, the game, draws what Core computes and sends orders into it, and nothing more. Core runs with no window, and it is being prepared for release as an open-source simulation tool for teaching, concept studies and research. It is not a validated operational tool.

12.1 What is Core and what is Live

PART	WHAT IT HOLDS	WHERE
Core: physical models	Two-body motion, Lambert, gravity with J2 and the moons, ephemerides, link budgets, antenna patterns, radar, heat and infrared, the heat forecast, sensing, the tracking filter, breakup and debris, civil traffic, and the world that ties them together	`godot/core/`
Core: scenario rules	Blue's effects and their costs, Red's behaviour, objectives and the debrief, the log's wording, save and continue. They run headless, but they are rules of play, and most of their values are Notional	`godot/core/`
Scenarios	Classes, equipment and constants; the current scenario's cast and objectives	`godot/data/*.json`
Live	The console, the map, the windows, art, sound, the tutorial	`godot/ui/`, `godot/scenes/`, `godot/assets/`

Table 6. The split. A check in the test suite fails if anything in Core names one of Live's classes.

12.2 Running Core without the game

A command-line runner loads a scenario, steps time and writes what happens as JSON lines, one object per line:

```
godot --headless --path godot --script res://cli/run.gd -- \
      --hours=24 --step=600 --seed=4417 --out=run.jsonl
```

The first line describes the run (scenario, seed, start time, units). Each step line carries the time, escalation, Blue's ships (position, velocity and Δv left), Blue's tracks with their 1σ ellipses, and every log entry since the last step. The last line is the debrief. Tracks report Blue's belief, never the truth; `--truth=true` adds the true states for analysis. Units are kilometres, km/s and seconds, centred on Jupiter. The test suite runs it on every change.

12.3 Real, Projected and Notional

Every model and value carries one of three labels, which extend the source labels of §1:

- **Real:** sourced and checked against an independent reference (§10). The orbits, gravity, the radio and radar equations, Planck's law, the filter.
- **Projected:** near-future and grounded in programmes that exist today, such as nuclear-thermal drives at NASA's tested 900 s specific impulse.
- **Notional:** invented for a scenario, the way defence and industry studies fly notional systems: Red's classes and behaviour, equipment sizes, escalation scores, the compressed timescale.

A Real model fed Notional inputs gives a Notional answer: the jamming equation is exact, but a Red jammer's power is a design value, and so is the range it yields. The project's fidelity table (`docs/FIDELITY.md`) gives each model's label, what it was checked against and what it leaves out.

12.4 Status of the release

As of this edition the code is still under an all-rights-reserved licence. Done: the asset licence audit, the Core and Live split with its check, the command-line runner, the fidelity table, and contribution, conduct and security guides. The jamming and counterspace models are built only from open, published sources, every one cited in this edition, and the owner has cleared them and the name for release. In the game, Blue's service is the fictional U.S. Space Corps, the name the House voted for in 2017, so the U.S. Space Force's name and seal never appear on screen; the sources still cite the real service by name. Still to do before release: choosing licences and a review of the repository's history. Contributions will be welcome once a licence is chosen; they must cite public sources only.

REFERENCES

Every source the game cites, in its code, data and tests, grouped by the model that uses it. A link opens the document itself, or a free copy where the publisher's is behind a paywall. Works with no readable copy online are listed without one.

EPHEMERIDES AND REFERENCE FRAMES

§2

1. JPL Solar System Dynamics Group. Horizons On-Line Ephemeris System. Jet Propulsion Laboratory, California Institute of Technology. JPL SSD ↗
2. Standish, E. M. & Williams, J. G. (1992). Keplerian Elements for Approximate Positions of the Major Planets. JPL Solar System Dynamics (web edition). JPL SSD ↗
3. Park, R. S., Folkner, W. M., Williams, J. G. & Boggs, D. H. (2021). The JPL Planetary and Lunar Ephemerides DE440 and DE441. Astronomical Journal 161(3):105. doi:10.3847/1538-3881/abd414 JPL SSD PDF ↗
4. JPL Solar System Dynamics. Planetary Satellite Mean Elements (Jupiter: JUP365). JPL SSD ↗
5. JPL Solar System Dynamics. Planetary Satellite Physical Parameters. JPL SSD ↗
6. Jacobson, R. A. (2021). The Orbits of the Regular Jovian Satellites and the Orientation of the Pole of Jupiter (JUP365). Personal communication to Horizons/NAIF. Corroborated by: JPL Solar System Dynamics, Planetary Satellite Mean Elements (JUP365), which gives this reference JPL SSD ↗ .
7. Archinal, B. A., Acton, C. H., A'Hearn, M. F. et al. (2018). Report of the IAU Working Group on Cartographic Coordinates and Rotational Elements: 2015. Celestial Mechanics and Dynamical Astronomy 130(3):22. doi:10.1007/s10569-017-9805-5 Corroborated by: NAIF/JPL (2022). Generic PCK kernel pck00011.tpc (IAU WGCCRE 2015 constants, citing Archinal et al. 2018) NAIF ↗ ; Archinal, B. A. et al. (2019). Correction to: Report of the IAU WGCCRE: 2015. CMDA 131:61 (DLR repository copy) ELIB.DLR.DE PDF ↗ .
8. IAU (2015). Resolution B3 on recommended nominal conversion constants for selected solar and planetary properties. XXIXth IAU General Assembly, Honolulu. See also Prša, A. et al. (2016). Nominal values for selected solar and planetary quantities: IAU 2015 Resolution B3. Astronomical Journal 152(2):41. IAU.ORG PDF ↗

9. Williams, D. R. Jupiter Fact Sheet. NASA Space Science Data Coordinated Archive (NSSDCA), Goddard Space Flight Center. NSSDC.GSFC.NASA.GOV ↗

10. Williams, D. R. Jovian Satellite Fact Sheet. NASA Space Science Data Coordinated Archive (NSSDCA), Goddard Space Flight Center. NSSDC.GSFC.NASA.GOV ↗

11. Peale, S. J. (1976). Orbital resonances in the solar system. Annual Review of Astronomy and Astrophysics 14:215–246. ADS SCAN PDF ↗ Corroborated by: Peale, S. J. & Lee, M. H. (2002). A primordial origin of the Laplace relation among the Galilean satellites. Science 298:593 (arXiv:astro-ph/0210589) ARXIV ↗ .

12. Peale, S. J., Cassen, P. & Reynolds, R. T. (1979). Melting of Io by tidal dissipation. Science 203(4383):892–894. Corroborated by: Morabito, L. A. (2012). Discovery of volcanic activity on Io: a historical review. arXiv:1211.2554 ARXIV ↗ .

GRAVITY AND TRAJECTORIES

§3

13. Folkner, W. M., Iess, L., Anderson, J. D. et al. (2017). Jupiter gravity field estimated from the first two Juno orbits. Geophysical Research Letters 44(10):4694–4700. doi:10.1002/2017GL073140 IRIS.UNIROMA1.IT PDF ↗

14. Iess, L., Folkner, W. M., Durante, D. et al. (2018). Measurement of Jupiter's asymmetric gravity field. Nature 555:220–222. doi:10.1038/nature25776 WEIZMANN.AC.IL PDF ↗

15. Durante, D., Parisi, M., Serra, D. et al. (2020). Jupiter's gravity field halfway through the Juno mission. Geophysical Research Letters 47(4):e2019GL086572. doi:10.1029/2019GL086572 CRIS.UNIBO.IT PDF ↗

16. Sicardy, B. (2010). Stability of the triangular Lagrange points beyond Gascheau's value. Celestial Mechanics and Dynamical Astronomy 107(1–2):145–155. doi:10.1007/s10569-010-9259-5 HAL ↗

17. Curtis, H. D. (2020). Orbital Mechanics for Engineering Students, 4th ed. Butterworth-Heinemann (Elsevier). Corroborated by: Weber, B. Orbital Mechanics & Astrodynamics (open textbook, CC BY-SA 4.0) OPEN TEXTBOOK ↗ ; Izzo, D. (2015). Revisiting Lambert's problem. CMDA 121:1-15 (arXiv:1403.2705) ARXIV ↗ ; NASA/TP-20220014814 (2022). Astrodynamics Convention and Modeling Reference for Lunar, Cislunar, and Libration Point Orbits. NASA NASA NTRS ↗ ; JPL SSD, Approximate Positions of the Planets (Standish & Williams 1992 web edition) JPL SSD ↗ .

18. Vallado, D. A. (2013). Fundamentals of Astrodynamics and Applications, 4th ed. Microcosm Press. Corroborated by: hapsira (open-source poliastro fork), core/propagation/vallado.py, documented as Vallado's KEPLER algorithm GITHUB ↗ ; NASA/TP-20220014814 (2022). Astrodynamics Convention and Modeling Reference for Lunar, Cislunar, and Libration Point Orbits. NASA NASA NTRS ↗ .

19. Bate, R. R., Mueller, D. D. & White, J. E. (1971). Fundamentals of Astrodynamics. Dover Publications. Corroborated by: Weber, B. Orbital Mechanics & Astrodynamics (open textbook, CC BY-SA 4.0) OPEN TEXTBOOK ↗ .

20. Montenbruck, O. & Gill, E. (2000). Satellite Orbits: Models, Methods and Applications. Springer. Corroborated by: NASA/TP-20220014814 (2022). Astrodynamics Convention and Modeling Reference for Lunar, Cislunar, and Libration Point Orbits. NASA NASA NTRS ↗ .

21. Aarseth, S. J. (2003). Gravitational N-Body Simulations: Tools and Algorithms. Cambridge University Press. Corroborated by: Pelupessy, F. I., Jaenes, J. & Portegies Zwart, S. (2012). N-body integrators with individual time steps from hierarchical splitting. New Astronomy 17:711 (arXiv:1205.5668) ARXIV ↗ .

22. Danby, J. M. A. (1987). The solution of Kepler's equation. III. Celestial Mechanics 40:303–312. ADS SCAN PDF ↗ Corroborated by: Zechmeister, M. (2018). CORDIC-like method for solving Kepler's equation. A&A 619:A128 (arXiv:1808.07062) ARXIV ↗ .

23. Izzo, D. (2015). Revisiting Lambert's problem. Celestial Mechanics and Dynamical Astronomy 121(1):1–15. doi:10.1007/s10569-014-9587-y ARXIV ↗

24. Broyden, C. G. (1965). A class of methods for solving nonlinear simultaneous equations. Mathematics of Computation 19(92):577–593. AMS.ORG PDF ↗ Corroborated by: Bindel, D. (2023). CS 4220 Numerical Analysis, lecture notes 2023-04-14 'Broyden'. Cornell University CS.CORNELL.EDU PDF ↗ .

25. Press, W. H., Teukolsky, S. A., Vetterling, W. T. & Flannery, B. P. (1992). Numerical Recipes in C: The Art of Scientific Computing, 2nd ed. Cambridge University Press. NR.COM READER ↗

26. Press, W. H., Teukolsky, S. A., Vetterling, W. T. & Flannery, B. P. (2007). Numerical Recipes: The Art of Scientific Computing, 3rd ed. Cambridge University Press. NUMERICAL.RECIPES ↗ Corroborated by: Bindel, D. (2023). CS 4220 lecture notes 'Broyden'. Cornell University CS.CORNELL.EDU PDF ↗ .

PROPULSION

§3

27. NASA Glenn Research Center. Ideal Rocket Equation. Beginner's Guide to Aeronautics (formerly Beginner's Guide to Rockets). GRC.NASA.GOV ↗

28. Tsiolkovsky, K. E. (1903). Exploration of Outer Space by Means of Reaction Devices [Issledovanie mirovykh prostranstv reaktivnymi priborami]. Nauchnoe Obozrenie, no. 5. English translation as 'A Rocket into Cosmic Space (1903)' in Works on Rocket Technology, NASA TT F-243 (1965). TSIOLKOVSKY.ORG PDF ↗

29. Sutton, G. P. & Biblarz, O. (2016). Rocket Propulsion Elements, 9th ed. Wiley. Corroborated by: Reed, B. D. (2004). On-board chemical propulsion technology. NASA/TM-2004-212698, NASA Glenn NASA NTRS ↗ .

30. Borowski, S. K., McCurdy, D. R. & Packard, T. W. (2009). Nuclear Thermal Rocket/Vehicle Characteristics and Sensitivity Trades for NASA's Mars Design Reference Architecture (DRA) 5.0 Study. Nuclear and Emerging Technologies for Space (NETS) 2009, Atlanta. NTRS 20120012928. NASA NTRS ↗

31. Borowski, S. K., McCurdy, D. R. & Packard, T. W. (2012). Nuclear Thermal Propulsion (NTP): A Proven Growth Technology for Human NEO/Mars Exploration Missions. IEEE Aerospace Conference, Big Sky, MT; NASA report E-18081, NTRS 20120003776. NASA NTRS ↗

32. NASA (2023, January 24). NASA, DARPA Will Test Nuclear Engine for Future Mars Missions [news release]. Demonstration Rocket for Agile Cislunar Operations (DRACO), a DARPA/NASA program. NASA.GOV ↗

33. ITU-R Recommendation SA.509-3 (12/2013). Space research earth station and radio astronomy reference antenna radiation pattern for use in interference calculations, including coordination procedures, for frequencies less than 30 GHz. ITU, Geneva. ITU PDF ↗

34. ITU-R Recommendation S.465-5 (1993). Reference earth-station radiation pattern for use in coordination and interference assessment in the frequency range from 2 to about 30 GHz. ITU, Geneva. (Superseded by S.465-6, 01/2010.) ITU PDF ↗

35. ITU-R Recommendation S.1428-1 (02/2001). Reference FSS earth-station radiation patterns for use in interference assessment involving non-GSO satellites in frequency bands between 10.7 GHz and 30 GHz. ITU, Geneva. ITU PDF ↗

36. ITU-R Recommendation S.672-4 (09/1997). Satellite antenna radiation pattern for use as a design objective in the fixed-satellite service employing geostationary satellites. ITU, Geneva. ITU PDF ↗

37. ITU-R Recommendation P.525-4 (08/2019). Calculation of free-space attenuation. ITU, Geneva. (Superseded by P.525-5, 11/2024.) ITU PDF ↗

38. ITU-R Recommendation SM.1055-0 (07/1994). The use of spread spectrum techniques. ITU, Geneva. ITU PDF ↗

39. JPL. DSN Telecommunications Link Design Handbook, 810-005: Module 201, Frequency and Channel Assignments, Rev. F (May 2022); Module 202, Doppler Tracking, Rev. D (Sept. 2021); Module 104, 34-m BWG Stations Telecommunications Interfaces, Rev. O (34 m X-band gain, Table 8; system temperatures, Table 11). Jet Propulsion Laboratory, Pasadena. JPL PDF ↗

40. CCSDS 401.0-B-32 (October 2021). Radio Frequency and Modulation Systems, Part 1: Earth Stations and Spacecraft. Recommended Standard (Blue Book), Issue 32. CCSDS, Washington, DC. CCSDS PDF ↗

41. IEEE Std 211-2018. IEEE Standard Definitions of Terms for Radio Wave Propagation. IEEE, New York (revision of IEEE Std 211-1997). Corroborated by: ITU-R P.372-17 (08/2024). Radio noise ITU PDF ↗.

42. Friis, H. T. (1946). A note on a simple transmission formula. Proceedings of the IRE 34(5):254-256. doi:10.1109/JRPROC.1946.234568 Corroborated by: Naval Air Warfare Center Weapons Division (2013). Electronic Warfare and Radar Systems Engineering Handbook, 4th ed., NAWCWD TP 8347 (DTIC ADA617071) EVERYSPEC.COM ↗.

43. Shannon, C. E. (1949). Communication in the presence of noise. Proceedings of the IRE 37(1):10-21. doi:10.1109/JRPROC.1949.232969 Corroborated by: Modiano, E. MIT OCW 16.36 Communication Systems Engineering (Spring 2009), Lecture 2 MIT OCW ↗.

44. Balanis, C. A. (2016). Antenna Theory: Analysis and Design, 4th ed. Wiley, Hoboken, NJ. Corroborated by: ITU-R P.525-4 (08/2019). Calculation of free-space attenuation ITU PDF ↗; Naval Air Warfare Center Weapons Division (2013). Electronic Warfare and Radar Systems Engineering Handbook, 4th ed., NAWCWD TP 8347 (DTIC ADA617071) EVERYSPEC.COM ↗.

45. Skolnik, M. I. (2001). Introduction to Radar Systems, 3rd ed. McGraw-Hill, New York.

46. Skolnik, M. I. (ed.) (2008). Radar Handbook, 3rd ed. McGraw-Hill, New York. Corroborated by: Naval Air Warfare Center Weapons Division (2013). Electronic Warfare and Radar Systems Engineering Handbook, 4th ed., NAWCWD TP 8347 (DTIC ADA617071) EVERYSPEC.COM ↗ ; ITU-R SA.509-3 (12/2013) ITU PDF ↗ .

47. Barton, D. K. (2005). Radar System Analysis and Modeling. Artech House, Norwood, MA. Corroborated by: MIT Lincoln Laboratory, Introduction to Radar Systems lecture series: Tracking and Parameter Estimation (2008) MIT LL ↗ .

48. Richards, M. A. (2005). Fundamentals of Radar Signal Processing. McGraw-Hill, New York.

49. Albersheim, W. J. (1981). A closed-form approximation to Robertson's detection characteristics. Proceedings of the IEEE 69(7):839. doi:10.1109/PROC.1981.12082 Corroborated by: Naval Air Warfare Center Weapons Division (2013). Electronic Warfare and Radar Systems Engineering Handbook, 4th ed., NAWCWD TP 8347 (DTIC ADA617071) EVERYSPEC.COM ↗ .

50. North, D. O. (1963). An analysis of the factors which determine signal/noise discrimination in pulsed-carrier systems. Proceedings of the IEEE 51(7):1016–1027 (reprint of RCA Laboratories Report PTR-6C, June 1943). doi:10.1109/PROC.1963.2383

51. Urkowitz, H. (1967). Energy detection of unknown deterministic signals. Proceedings of the IEEE 55(4):523–531. doi:10.1109/PROC.1967.5573 Corroborated by: Rashid, M. & Nanzer, J. A. (2024). Expectation Maximization Aided Modified Weighted Sequential Energy Detector for Distributed Cooperative Spectrum Sensing. arXiv:2403.19556 ARXIV PDF ↗ ; Condon, J. J. & Ransom, S. M. Essential Radio Astronomy, ch. 3 (NRAO / Princeton Univ. Press 2016, online edition) NRAO ↗ .

52. Dicke, R. H. (1946). The measurement of thermal radiation at microwave frequencies. Review of Scientific Instruments 17(7):268–275. doi:10.1063/1.1770483 Corroborated by: Condon, J. J. & Ransom, S. M. Essential Radio Astronomy, ch. 3 (NRAO / Princeton Univ. Press 2016, online edition) NRAO ↗ .

53. Wilson, T. L., Rohlfs, K. & Hüttemeister, S. (2013). Tools of Radio Astronomy, 6th ed. Springer, Berlin. Corroborated by: Condon, J. J. & Ransom, S. M. Essential Radio Astronomy, ch. 3 (NRAO / Princeton Univ. Press 2016, online edition) NRAO ↗ .

54. Naval Air Warfare Center Weapons Division (2013). Electronic Warfare and Radar Systems Engineering Handbook, 4th ed., NAWCWD TP 8347. Point Mugu, CA. (DTIC ADA617071, approved for public release.) DTIC PDF ↗

55. Kerczewski, R. J., Yu, J., Walton, J. C., Perl, T. D., Andro, M. & Alexovich, R. E. (1992). Multiple-access phased array antenna simulator for a digital beam forming system investigation. NASA TM-105422 (Ninth International Conference on Digital Satellite Communications, Copenhagen, May 1992). NASA NTRS PDF ↗

56. Gardner, W. A. (1991). Exploitation of spectral redundancy in cyclostationary signals. IEEE Signal Processing Magazine 8(2):14–36. doi:10.1109/79.81007 CYCLOSTATIONARITY.COM PDF ↗

57. Self, A. G. & Smith, B. G. (1985). Intercept time and its prediction. IEE Proceedings F (Communications, Radar and Signal Processing) 132(4):215. doi:10.1049/ip-f-1.1985.0052

58. Clarkson, I. V. L., Perkins, J. E. & Mareels, I. M. Y. (1996). Number/theoretic solutions to intercept time problems. IEEE Transactions on Information Theory 42(3):959–971. doi:10.1109/18.490557

59. Wiley, R. G. (2006). ELINT: The Interception and Analysis of Radar Signals. Artech House, Norwood, MA.

60. Pace, P. E. (2009). Detecting and Classifying Low Probability of Intercept Radar, 2nd ed. Artech House, Norwood, MA.

61. Torrieri, D. (2022). Principles of Spread-Spectrum Communication Systems, 5th ed. Springer, Cham.

62. Donkersley, J. (with Rouma, A.) (2022). SDR++ User Guide, for SDR++ up to version 1.1. SDR++ software by Alexandre Rouma. SDRPP.ORG PDF ↗

63. GNU Radio Foundation / SigMF project. Signal Metadata Format (SigMF) Specification, version 1.2.x (v1.2.6 current). SIGMF.GITHUB.IO ↗

64. AMD (2023). Zynq UltraScale+ RFSoC Data Sheet: Overview, DS889 (v1.14), 27 June 2023. DOCS.AMD.COM ↗

DIRECTION FINDING AND GEOLOCATION §4

65. Knapp, C. H. & Carter, G. C. (1976). The generalized correlation method for estimation of time delay. IEEE Transactions on Acoustics, Speech, and Signal Processing 24(4):320-327. doi:10.1109/TASSP.1976.1162830 Corroborated by: Obo, H. et al. (2025). Noise Suppression for Time Difference of Arrival: Performance Evaluation of a Generalized Cross-Correlation Method Using Mean Signal and Inverse Filter. arXiv:2512.05355 ARXIV PDF ↗ .

66. Thompson, A. R., Moran, J. M. & Swenson, G. W., Jr. (2017). Interferometry and Synthesis in Radio Astronomy, 3rd ed. Astronomy and Astrophysics Library, Springer, Cham. doi:10.1007/978-3-319-44431-4 SPRINGER (OPEN) ↗

67. Poisel, R. A. (2012). Electronic Warfare Target Location Methods, 2nd ed. Artech House, Norwood, MA. Corroborated by: Thompson, A. R., Moran, J. M. & Swenson, G. W. (2017). Interferometry and Synthesis in Radio Astronomy, 3rd ed., Springer (open access) SPRINGER (OPEN) PDF ↗ ; Herath, S. C. & Pathirana, P. N. (2013). Optimal sensor arrangements in Angle of Arrival (AoA) and range based localization with linear sensor arrays. Sensors 13(9):12277-12294. doi:10.3390/s130912277 (open access, PMC3821368) PMC (OPEN) ↗ .

68. Stein, S. (1981). Algorithms for ambiguity function processing. IEEE Transactions on Acoustics, Speech, and Signal Processing 29(3):588-599. doi:10.1109/TASSP.1981.1163621 Corroborated by: Fowler, M. L. EECE 522 Notes 15: ML for TDOA/FDOA (revised). Binghamton University course notes WS2.BINGHAMTON.EDU PDF ↗ ; Obo, H. et al. (2025), arXiv:2512.05355 (Knapp & Carter coherence CRLB) ARXIV PDF ↗ .

69. Ho, K. C. & Chan, Y. T. (1997). Geolocation of a known altitude object from TDOA and FDOA measurements. IEEE Transactions on Aerospace and Electronic Systems 33(3):770-783. doi:10.1109/7.599239

70. Ho, K. C. & Xu, W. (2004). An accurate algebraic solution for moving source location using TDOA and FDOA measurements. IEEE Transactions on Signal Processing 52(9):2453-2463. doi:10.1109/TSP.2004.831921

71. Ho, K. C., Lu, X. & Kovavisaruch, L. (2007). Source localization using TDOA and FDOA measurements in the presence of receiver location errors: analysis and solution. IEEE Transactions on Signal Processing 55(2):684-696. doi:10.1109/TSP.2006.885744

72. Sarda, K., Roth, N., Zee, R. E., CaJacob, D. & Orr, N. G. (2018). Making the invisible visible: precision RF-emitter geolocation from space by the HawkEye 360 Pathfinder mission. 32nd Annual AIAA/USU Conference on Small Satellites, Logan, UT, paper SSC18-II-06. (Also presented at IAC 2018 as IAC-18-B4.4.1.) SMALLSAT 2018 PDF ↗

73. ESA eoPortal. HawkEye 360 (satellite mission page). EOPORTAL.ORG ↗

74. Burt, E. A., Prestage, J. D., Tjoelker, R. L. et al. (2021). Demonstration of a trapped-ion atomic clock in space. Nature 595:43–47. doi:10.1038/s41586-021-03571-7

ELECTROMAGNETIC ATTACK AND DIRECTED ENERGY §5

75. Adamy, D. L. (2001). EW 101: A First Course in Electronic Warfare; (2004) EW 102: A Second Course in Electronic Warfare; (2009) EW 103: Tactical Battlefield Communications Electronic Warfare. Norwood, MA: Artech House. Corroborated by: Naval Air Warfare Center Weapons Division (2013). Electronic Warfare and Radar Systems Engineering Handbook, NAWCWD TP 8347, 'Support Jamming' section (public domain; RF Cafe reproduction) RFCAFE.COM ↗ .

76. Cummings, W. C. (1992). An adaptive nulling antenna for military satellite communications. The Lincoln Laboratory Journal 5(2):173–194. MIT LL PDF ↗

77. Sabharwal, A., Schniter, P., Guo, D., Bliss, D. W., Rangarajan, S. & Wichman, R. (2014). In-band full-duplex wireless: challenges and opportunities. IEEE Journal on Selected Areas in Communications 32(9):1637–1652. doi:10.1109/JSAC.2014.2330193 ARXIV ↗

78. Danev, B., Zanetti, D. & Čapkun, S. (2012). On physical-layer identification of wireless devices. ACM Computing Surveys 45(1), article 6:1–29. doi:10.1145/2379776.2379782

79. Riyaz, S., Sankhe, K., Ioannidis, S. & Chowdhury, K. (2018). Deep learning convolutional neural networks for radio identification. IEEE Communications Magazine 56(9):146–152. doi:10.1109/MCOM.2018.1800153

80. Merchant, K., Revay, S., Stantchev, G. & Nousain, B. (2018). Deep learning for RF device fingerprinting in cognitive communication networks. IEEE Journal of Selected Topics in Signal Processing 12(1):160–167. doi:10.1109/JSTSP.2018.2796446

81. Kaushal, H. & Kaddoum, G. (2017). Optical communication in space: challenges and mitigation techniques. IEEE Communications Surveys & Tutorials 19(1):57–96. doi:10.1109/COMST.2016.2603518 ARXIV PREPRINT ↗

82. Klein, B. J. & Degnan, J. J. (1974). Optical antenna gain. 1: Transmitting antennas. Applied Optics 13(9):2134–2141. doi:10.1364/AO.13.002134 Corroborated by: Safi, H., Wang, Z., Mast, S., Haas, H. & Tavakkolnia, I. (2026). Analytical channel modeling and stability aware optimization of optical inter-satellite links. arXiv:2609.17431 ARXIV ↗ .

83. Papoulis, A. & Pillai, S. U. (2002). Probability, Random Variables and Stochastic Processes, 4th ed. New York: McGraw-Hill.

84. Benford, J., Swegle, J. A. & Schamiloglu, E. (2016). High Power Microwaves, 3rd ed. Boca Raton: CRC Press.

85. Zervas, M. N. & Codemard, C. A. (2014). High power fiber lasers: a review. IEEE Journal of Selected Topics in Quantum Electronics 20(5):219–241 (article 0904123). doi:10.1109/JSTQE.2014.2321279 Corroborated by: Hecht, J. (13 Apr 2016). Photonic Frontiers: High-efficiency optical pumping. Laser Focus World 52(4) LASERFOCUSWORLD.COM ↗ .

86. Lockheed Martin (28 Jan 2014). Lockheed Martin Demonstrates Weapons-Grade High Power Fiber Laser [press release]. Bothell, WA. NEWS.LOCKHEEDMARTIN.COM ↗

87. Hecht, J. (13 Apr 2016). Photonic Frontiers: High-efficiency optical pumping: 'Going green' cranks up the laser power. Laser Focus World 52(4). LASERFOCUSWORLD.COM ↗

88. Lockheed Martin (18 Aug 2022). Lockheed Martin Delivers Integrated Multi-Mission Laser Weapon System to the Navy [press release on HELIOS, 60+ kW class]. Related: Lockheed Martin (28 Jul 2023), Lockheed Martin to Scale Its Highest Powered Laser to 500 Kilowatts Power Level (OUSD(R&E) High Energy Laser Scaling Initiative, HELSI). NEWS.LOCKHEEDMARTIN.COM ↗

89. Air Force Research Laboratory (30 Jul 2021). AFRL's drone killer, THOR will welcome new drone 'hammer' [news release on THOR and the Mjolnir follow-on]. CHAMP: Boeing/AFRL flight test, Utah Test and Training Range, 16 Oct 2012. AFRESEARCHLAB.COM ↗

90. Missile Defense Agency (Jan 2024). Ground-based Midcourse Defense [fact sheet, 24-MDA-11677]. MDA.MIL PDF ↗

91. L3Harris Technologies (Dec 2025). L3Harris Delivers First Meadowlands Production Unit to US Space Force [news release; Meadowlands is the upgraded Counter Communications System]. L3HARRIS.COM ↗

92. Albon, C. (15 Apr 2025). Space Force 'Meadowlands' jammer ready for formal testing. Defense News. DEFENSENEWS.COM ↗

93. NASA (2023). DSOC (Deep Space Optical Communications) Technology Demonstration [fact sheet, dated 15 Jun 2023]. NASA/JPL-Caltech. NASA.GOV PDF ↗

94. Speed, R. (30 Sep 2025). NASA's deep-space laser comms demo has left the chat. The Register. THEREGISTER.COM ↗

POWER AND HEAT

§6

95. Henninger, J. H. (1984). Solar Absorptance and Thermal Emittance of Some Common Spacecraft Thermal-Control Coatings. NASA Reference Publication 1121, NASA Goddard Space Flight Center. NASA NTRS PDF ↗

96. Mason, L. S. (2003). A Power Conversion Concept for the Jupiter Icy Moons Orbiter. NASA/TM-2003-212596 (AIAA 2003-6007, 1st International Energy Conversion Engineering Conference), NASA Glenn Research Center. NASA NTRS PDF ↗

97. Mason, L. S. (2004). Power-Conversion Concept Designed for the Jupiter Icy Moons Orbiter. In Research and Technology 2003, NASA Glenn Research Center. NTRS 20050192432. NASA NTRS PDF ↗

98. Siamidis, J., Mason, L., Beach, D. & Yuko, J. (2005). Heat Rejection Concepts for Brayton Power Conversion Systems. NASA/TM-2005-213337 (AIAA 2004-5654, 2nd International Energy Conversion Engineering Conference). NASA NTRS PDF ↗

99. Poston, D. I., Gibson, M. A., Sanchez, R. G. & McClure, P. R. (2020). Results of the KRUSTY Nuclear System Test. Nuclear Technology 206(sup1):S89-S117. doi:10.1080/00295450.2020.1730673 OSTI PDF ↗

100. Choi, M. K. (2012). Phase Change Material for Temperature Control of Loop Heat Pipe Compensation Chamber. 10th International Energy Conversion Engineering Conference (AIAA), NASA GSFC. NTRS 20120015405. NASA NTRS PDF ↗

101. Choi, M. K. (2018). Thermal Assessment of Paraffin Phase Change Material Mini-Packs on IceCube 3U CubeSat in Flight. 2018 AIAA Propulsion and Energy Forum, NASA GSFC. NTRS 20180004470. NASA NTRS PDF ↗

102. Leimkuehler, T. O., Stephan, R. A. & Hawkins-Reynolds, E. (2011). Testing and Failure Mechanisms of Ice Phase Change Material Heat Exchangers. 41st International Conference on Environmental Systems (AIAA), NASA JSC. NTRS 20110008723. NASA NTRS PDF ↗

103. Gilmore, D. G. (ed.) (2002). Spacecraft Thermal Control Handbook, Vol. 1: Fundamental Technologies, 2nd ed. The Aerospace Press / AIAA, El Segundo, CA. Corroborated by: Gaite, J. & Fernández-Rico, G. (2012). Linear approach to the orbiting spacecraft thermal problem. Journal of Thermophysics and Heat Transfer 26:511-522. arXiv:1202.4372 ARXIV ↗ .

104. Incropera, F. P., DeWitt, D. P., Bergman, T. L. & Lavine, A. S. (2007). Fundamentals of Heat and Mass Transfer, 6th ed. Wiley. Corroborated by: Çengel, Y. A. & Ghajar, A. J. Heat and Mass Transfer: Fundamentals and Applications, Appendix 1, Table A-3 'Properties of solid metals' (free appendix PDF from McGraw-Hill) HIGHERED.MHEDUCATION.COM PDF ↗ .

105. Chase, M. W., Jr. (1998). NIST-JANAF Thermochemical Tables, 4th ed., J. Phys. Chem. Ref. Data Monograph 9; as tabulated in the NIST Chemistry WebBook, SRD 69: Aluminum, solid-phase Shomate equation, 298–933 K (Cp = 24.21 J/mol·K at 298.15 K, 897 J/kg·K). NIST WEBBOOK ↗

106. Cox, A. N. (ed.) (2000). Allen's Astrophysical Quantities, 4th ed. AIP Press / Springer, New York. Corroborated by: Tripathi, A. K., Chaubey, S. & Singh, A. K. (2020). Study of Limb Darkening Effect and Rotation Period of Sun by using Solar Telescope. Journal of Scientific Research (Banaras Hindu University) 64(1). doi:10.37398/JSR.2020.640132 BHU.AC.IN PDF ↗ .

107. Hanel, R. A., Conrath, B. J., Herath, L. W., Kunde, V. G. & Pirraglia, J. A. (1981). Albedo, internal heat, and energy balance of Jupiter: Preliminary results of the Voyager infrared investigation. Journal of Geophysical Research 86(A10):8705-8712. doi:10.1029/JA086iA10p08705 NASA NTRS PDF ↗

108. Li, L., Jiang, X., West, R. A., Gierasch, P. J., Perez-Hoyos, S., Sanchez-Lavega, A., Fletcher, L. N., Fortney, J. J. et al. (2018). Less absorbed solar energy and more internal heat for Jupiter. Nature Communications 9:3709. doi:10.1038/s41467-018-06107-2 NATURE (OPEN) ↗

109. Kopp, G. & Lean, J. L. (2011). A new, lower value of total solar irradiance: Evidence and climate significance. Geophysical Research Letters 38:L01706. doi:10.1029/2010GL045777 AGU (OPEN) ↗

110. Edwards, D. (2012). An Overview of the Jupiter Icy Moons Orbiter (JIMO) Mission, Environments, and Materials Challenges. NASA MSFC presentation, 12th International Symposium on Materials in the Space Environment. NTRS 20120016895. [Programme: NASA Project Prometheus, JIMO, 2003-2005.] NASA NTRS PDF ↗

INFRARED AND OPTICAL SENSING

§6

111. Planck, M. (1901). Ueber das Gesetz der Energieverteilung im Normalspectrum. Annalen der Physik 309(3):553-563 (4. Folge, Bd. 4). doi:10.1002/andp.19013090310 WIKISOURCE ↗

112. Widger, W. K. & Woodall, M. P. (1976). Integration of the Planck blackbody radiation function. Bulletin of the American Meteorological Society 57(10):1217-1219. doi:10.1175/1520-0477(1976)057<1217:IOTPBR>2.0.CO;2 AMS (OPEN) ↗ Corroborated by: Lewis, W. & McClarren, R. G. (2025). Fast, Accurate Numerical Evaluation of Incomplete Planck Integrals. arXiv:2503.08892 (University of Notre Dame) ARXIV ↗ .

113. Hudson, R. D. (1969). Infrared System Engineering. Wiley, New York. Corroborated by: OpenStax, University Physics Volume 2, §1.6 Mechanisms of Heat Transfer (Rice University, CC BY) OPENSTAX ↗ .

114. Hecht, E. (2017). Optics, 5th ed. Pearson. Corroborated by: OpenStax, University Physics Volume 3, §4.5 Circular Apertures and Resolution, Eq. (4.5) OPENSTAX ↗ ; Barbastathis, G. MIT OCW 2.71 Optics (Spring 2009), Lecture 17: Fraunhofer diffraction; Fourier transforms and theorems MIT OCW ↗ .

115. Mainzer, A. K., Masiero, J. R., Abell, P. A., Bauer, J. M., Bottke, W. et al. (2023). The Near-Earth Object Surveyor Mission. Planetary Science Journal 4(12):224. doi:10.3847/PSJ/ad0468 IOP (OPEN) ↗

116. NASA Jet Propulsion Laboratory. Near-Earth Object Surveyor (mission page). Accessed 2026. JPL.NASA.GOV ↗

117. Matsumoto, T., Tsumura, K., Matsuoka, Y. & Pyo, J. (2018). Zodiacal light beyond Earth orbit observed with Pioneer 10. Astronomical Journal 156(3):86. doi:10.3847/1538-3881/aad0f0 ARXIV ↗

118. Leinert, Ch., Bowyer, S., Haikala, L. K., Hanner, M. S., Hauser, M. G., Levasseur-Regourd, A.-Ch., Mann, I., Mattila, K. et al. (1998). The 1997 reference of diffuse night sky brightness. Astronomy and Astrophysics Supplement Series 127:1-99. doi:10.1051/aas:1998105 AAS.AANDA.ORG PDF ↗

119. Stokes, G. H., Weber, R., Shelly, F. C., Beatty, D. E., Viggh, H. E. M., Rork, E. W. & Hays, B. (1996). Air Force Planetary Defense System: Initial Field Test Results. In Space 96: Proceedings of the Fifth International Conference on Space, Engineering, Construction, and Operations in Space V (ASCE), Albuquerque, 1-6 June 1996, pp. 46-53. doi:10.1061/40177(207)7 Corroborated by: Stokes, G. H., Evans, J. B., Viggh, H. E. M., Shelly, F. C. & Pearce, E. C. (1998). The Lincoln Near-Earth Asteroid Research (LINEAR) Program. Lincoln Laboratory Journal 11(1):27-40 MIT LL PDF ↗ .

120. U.S. Space Force. Ground-Based Electro-Optical Deep Space Surveillance (GEODSS) fact sheet (sites: Socorro NM, Maui HI, Diego Garcia). SPACEFORCE.MIL ↗

121. Johnson, J. (1958). Analysis of image forming systems. In Proceedings of the Image Intensifier Symposium, 6-7 October 1958, U.S. Army Engineer Research and Development Laboratories, Fort Belvoir, VA, pp. 244-273

(DTIC AD220160). Corroborated by: Sjaardema, T. A., Smith, C. S. & Birch, G. C. (2015). History and Evolution of the Johnson Criteria. Sandia National Laboratories, SAND2015-6368. OSTI 1222446 OSTI PDF ↗ .

122. Madhusudhan, N. & Burrows, A. (2012). Analytic models for albedos, phase curves, and polarization of reflected light from exoplanets. Astrophysical Journal 747(1):25. doi:10.1088/0004-637X/747/1/25 ARXIV ↗

123. Willmer, C. N. A. (2018). The absolute magnitude of the Sun in several filters. Astrophysical Journal Supplement Series 236(2):47. doi:10.3847/1538-4365/aabfdf ARXIV ↗

124. Torres, G. (2010). On the use of empirical bolometric corrections for stars. Astronomical Journal 140(5):1158–1162. doi:10.1088/0004-6256/140/5/1158 ARXIV ↗

TRACKING

§7

125. Julier, S. J. & Uhlmann, J. K. (2004). Unscented filtering and nonlinear estimation. Proceedings of the IEEE 92(3):401–422. doi:10.1109/JPROC.2003.823141 Corroborated by: filterpy 1.4.5 documentation, JulierSigmaPoints (R. Labbe) FILTERPY.READTHEDOCS.IO ↗ .

126. Bar-Shalom, Y., Li, X. R. & Kirubarajan, T. (2001). Estimation with Applications to Tracking and Navigation. New York: Wiley. Corroborated by: Labbe, R. R. Kalman and Bayesian Filters in Python (open textbook, CC-BY 4.0) GITHUB ↗ ; Labbe, R. R. Kalman and Bayesian Filters in Python, ch. 14 'Adaptive Filtering' (open textbook) GITHUB ↗ .

127. Lindgren, A. G. & Gong, K. F. (1978). Position and velocity estimation via bearing observations. IEEE Transactions on Aerospace and Electronic Systems AES-14(4):564–577. doi:10.1109/TAES.1978.308681 Corroborated by: Bu, S., Meng, A. & Zhou, G. (2021). A New Pseudolinear Filter for Bearings-Only Tracking without Requirement of Bias Compensation. Sensors 21(16):5444 PMC (OPEN) ↗ .

128. Aidala, V. J. & Nardone, S. C. (1982). Biased estimation properties of the pseudolinear tracking filter. IEEE Transactions on Aerospace and Electronic Systems AES-18(4):432–441. doi:10.1109/TAES.1982.309250 Corroborated by: Bu, S., Meng, A. & Zhou, G. (2021). Sensors 21(16):5444 PMC (OPEN) ↗ .

129. Woffinden, D. C. & Geller, D. K. (2009). Observability criteria for angles-only navigation. IEEE Transactions on Aerospace and Electronic Systems 45(3):1194–1208. doi:10.1109/TAES.2009.5259193

130. Blackman, S. & Popoli, R. (1999). Design and Analysis of Modern Tracking Systems. Norwood, MA: Artech House. Corroborated by: MathWorks. trackHistoryLogic (Sensor Fusion and Tracking Toolbox documentation), citing Blackman & Popoli 1999 MATHWORKS.COM ↗ .

DEBRIS AND BREAKUP

§5

131. Johnson, N. L., Krisko, P. H., Liou, J.-C. & Anz-Meador, P. D. (2001). NASA's new breakup model of EVOLVE 4.0. Advances in Space Research 28(9):1377–1384. doi:10.1016/S0273-1177(01)00423-9 Corroborated by: Frey, S. & Colombo, C. (2021). Transformation of satellite breakup distribution for probabilistic orbital collision hazard analysis. JGCD 44(1):88-105 (PoliMi IRIS open-access copy) RE.PUBLIC.POLIMI.IT PDF ↗ ; Krisko, P. H. (2011). Proper implementation of the 1998 NASA breakup model. ODQN 15(4):4-5 NASA ODPO PDF ↗ ;

Panarotto, S. (2024, a.y. 2023/24). Analisi del delta V di detriti spaziali generati a seguito di eventi di frammentazione in orbita. MSc thesis, Univ. Padova THESIS.UNIPD.IT PDF ↗ .

132. Krisko, P. H. (2011). Proper implementation of the 1998 NASA breakup model. Orbital Debris Quarterly News 15(4):4–5. NASA ODPO PDF ↗

133. Joubert, W. & Tingay, S. (2020). Simulations of orbital debris clouds due to breakup events and their characterisation using the Murchison Widefield Array radio telescope. Experimental Astronomy 51(1):61–75 (issue dated 2021). doi:10.1007/s10686-020-09684-7; arXiv:2008.03657 ARXIV ↗

134. Thiele, S. & Boley, A. C. (2022). Investigating the risks of debris-generating ASAT tests in the presence of megaconstellations. The Journal of the Astronautical Sciences 69(6):1797–1820. doi:10.1007/s40295-022-00356-6; arXiv:2111.12196 ARXIV ↗

135. Frey, S. & Colombo, C. (2021). Transformation of satellite breakup distribution for probabilistic orbital collision hazard analysis. Journal of Guidance, Control, and Dynamics 44(1):88–105. doi:10.2514/1.G004939. Appendix B: the NASA standard breakup model as implemented. RE.PUBLIC.POLIMI.IT PDF ↗

136. Panarotto, S. (2024; academic year 2023/24). Analisi del delta V di detriti spaziali generati a seguito di eventi di frammentazione in orbita [Analysis of the delta-V of space debris generated by in-orbit fragmentation events]. MSc thesis (Laurea Magistrale, Ingegneria Aerospaziale), Università degli Studi di Padova; supervisor L. Olivieri. THESIS.UNIPD.IT PDF ↗

137. NASA Orbital Debris Program Office (2007). Chinese anti-satellite test creates most severe orbital debris cloud in history. Orbital Debris Quarterly News 11(2):2–3. And: (2022). The intentional destruction of Cosmos 1408. Orbital Debris Quarterly News 26(1):1–4. NASA ODPO PDF ↗

138. Anz-Meador, P. D., Opiela, J. N., Shoots, D. & Liou, J.-C. (2018). History of On-orbit Satellite Fragmentations, 15th ed. NASA/TM-2018-220037, NASA JSC Orbital Debris Program Office. NASA NTRS PDF ↗

DOCTRINE

§8

139. Joint Chiefs of Staff (2020). JP 3-85, Joint Electromagnetic Spectrum Operations, 22 May 2020. FAS PDF ↗

140. Joint Chiefs of Staff (2018, incorporating Change 1, 26 Oct 2020). JP 3-14, Space Operations, 10 April 2018. FAS PDF ↗

141. Joint Chiefs of Staff (2018). JP 3-60, Joint Targeting, 28 September 2018. ESD.WHS.MIL PDF ↗

142. U.S. Air Force, LeMay Center (2026). AFDP 3-60, Targeting, 1 May 2026. DOCTRINE.AF.MIL PDF ↗

143. U.S. Space Force (2025). Space Warfighting: A Framework for Planners. March 2025 (released April 2025). CSPS.AEROSPACE.ORG PDF ↗

144. U.S. Space Force (2025). Space Force Doctrine Document 1 (SFDD-1), The Space Force, signed 3 April 2025. SPACEFORCE.MIL PDF ↗

145. Department of Defense (2014). MIL-STD-2525D, Department of Defense Interface Standard: Joint Military Symbology, 10 June 2014. JCS.MIL PDF ↗ Corroborated by: milsymbol (spatialillusions), MIL-STD-2525D implementation test page reproducing the standard's tables CDN.JSDELIVR.NET ↗ .

THE SETTING AND THE SCENARIO §9

146. U.S. Space Force, Combat Forces Command (formerly Space Operations Command). Fact sheet: Mission Delta 2 - Space Domain Awareness (companion fact sheets: Delta 3 - Space Electromagnetic Warfare; Delta 9 - Orbital Warfare). USSF-CFC.SPACEFORCE.MIL ↗

147. Combined Space Operations (CSpO) Initiative (2022). Combined Space Operations Vision 2031. Released February 2022. DEFENCE.GOVT.NZ PDF ↗

148. NATO. NATO's approach to space (official topic page; documents the December 2019 London Declaration recognising space as an operational domain and the October 2020 decision to establish a NATO Space Operations Centre at Allied Air Command, Ramstein). NATO.INT ↗

149. UK Ministry of Defence (2021). UK Space Command (GOV.UK guidance page, published 1 April 2021; withdrawn 25 June 2026). GOV.UK ↗

150. U.S. Space Force. Fact sheet: Geosynchronous Space Situational Awareness Program (GSSAP) (current as of Oct 2020). SPACEFORCE.MIL ↗

151. National Reconnaissance Office. SILENTBARKER/NROL-107 launch page (launched 10 September 2023; joint NRO-U.S. Space Force space domain awareness mission). NRO.GOV ↗

152. U.S. Space Force / SAF Public Affairs (2 December 2023). US, UK, Australia announce trilateral Deep Space Advanced Radar Capability initiative. SPACEFORCE.MIL ↗

153. Defense Intelligence Agency (2022). Challenges to Security in Space 2022 (follow-up to the February 2019 edition). DIA.MIL PDF ↗ Corroborated by: Joint Chiefs of Staff news (11 Feb 2019), 'Defense Intelligence Agency releases report on challenges to U.S. security in space' JCS.MIL ↗ ; Weeden, B. & Samson, V. (eds.) (2023). Global Counterspace Capabilities, Secure World Foundation SWF PDF ↗ .

154. Strout, N., Defense News (23 July 2020). Russia conducted anti-satellite test in space, says US Space Command. Reports the USSPACECOM statement that Cosmos 2543 released an object in a non-destructive on-orbit ASAT test on 15 July 2020. DEFENSENEWS.COM ↗

155. Swope, C., Bingen, K. A., Young, M., Chang, M., Songer, S. & Tammelleo, J. (2024). Space Threat Assessment 2024. CSIS Aerospace Security Project, April 2024. AEROSPACE.CSIS.ORG PDF ↗

156. Weeden, B. & Samson, V. (eds.) (2023). Global Counterspace Capabilities: An Open Source Assessment. Secure World Foundation. Co-orbital inspection, including Luch/Olymp-K at GEO from 2014, Cosmos 2542/2543 shadowing USA 245 (2019–2020) and Shijian-21 towing Compass-G2 (2022). SWF PDF ↗

157. Xinhua, via China Aerospace Science and Technology Corporation (20 March 2024). China launches new relay satellite for Earth-Moon communications (Queqiao-2). The original Queqiao (2018) operates in an Earth-Moon L2 halo orbit. ENGLISH.SPACECHINA.COM ↗

158. Hitchens, T., Breaking Defense (19 October 2026). Russian satellite stalked ICEYE spacecraft in potentially dangerous on-orbit maneuvers. BREAKINGDEFENSE.COM ↗

159. Trevithick, J. & Altman, H., The War Zone (19 October 2026). Russian Satellite's Unprecedented Shadowing Of A Very Particular Radar Imaging Satellite Is A Big Deal. TWZ.COM ↗

160. ESA Earth Online. ICEYE (mission description: X-band SAR constellation of sub-100 kg satellites). EARTH.ESA.INT ↗

161. NASA/JPL (16 June 2016). NASA's Juno Spacecraft to Risk Jupiter's Fireworks for Science. News release. JPL.NASA.GOV ↗

162. Harwood, W., CBS News (2 Jul 2010). Progress docking aborted: Progress M-06M lost its telemetry link on final approach to the ISS and flew past. CBSNEWS.COM ↗

UNITS AND CONSTANTS

163. BIPM (2019). The International System of Units (SI), 9th edition (SI Brochure). BIPM.ORG PDF ↗

164. Tiesinga, E., Mohr, P. J., Newell, D. B. & Taylor, B. N. (2021). CODATA recommended values of the fundamental physical constants: 2018. Rev. Mod. Phys. 93, 025010. PHYSICS.NIST.GOV PDF ↗

165. Thompson, A. & Taylor, B. N. (2008). Guide for the Use of the International System of Units (SI). NIST Special Publication 811, 2008 edition. NVLPUBS.NIST.GOV PDF ↗

© 2026 Kevin Williamson. All rights reserved. Planetary data from NASA and JPL, used as published; NASA and JPL have not reviewed or endorsed this game. All ships, operators and events in the game are original fiction.